

Proof companion

Coupling-gated cortical coherence

A companion to the repository's Lean development

Second coverage stage: 17 September 2026

Abstract

This document presents mathematical explanations of 129 selected Lean results from 7 source modules. It places the assumptions and the argument of each result beside an appendix generated from Lean's elaborated statements. The saved extraction also contains 4973 nested tactic records, their before/after goal states, proof dependencies, and the fields of project structures reached through statement types. The mathematical prose is an editorial draft checked against the source during preparation; Lean verifies the formal proofs, not this prose.

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1 Reading and auditing the companion

The explained arguments are of six kinds: exact and approximate gluing, the critical exponent of stationary coherent solutions, the conditional composition of the framework, the witnesses and counterexamples that fence that composition, a no-go for the vertex-supported kernel beside the cell-pair construction that avoids it, and a concrete nonconstant family. The selection is explicit in `selection.json`. Each of its declarations has an extracted statement in the appendix, including implicit type and instance parameters. Definitions and hypotheses abbreviated in the mathematics are spelled out where they carry substantive content.

The selection follows a stated rule rather than a list of favourites. Three modules --- the chain and the two analytic results it consumes --- are covered *completely*: every theorem they declare has an extracted statement here, so a reader can see what those modules contain and not only what the prose chose to discuss. The finite-measure gluing module is covered completely on the same rule. Three further modules contribute selected witnesses and convergence results. Coverage of the rest of the development remains selected, and the coverage table records exactly which declarations are present.

The prose groups rewrites and elementary algebra into mathematical steps. Full tactic histories remain in the JSON snapshots under `data/`; Section 7.1 displays a small actual trace. A

nested tactic record is an elaboration event, so its count includes wrappers and macro expansions. It is not a count of independent mathematical steps.

Three distinctions guide the reading. A theorem can depend on physical conditions supplied as structure fields. A formally valid conditional theorem can have hypotheses that require separate witnesses. A dependency on a named definition does not establish an empirical interpretation of that definition. The examples below keep these distinctions visible.

The extractor rejects elaboration errors and checks that each selected result's axioms are drawn only from `propext`, `Classical.choice`, and `Quot.sound`. This is a check of logical dependencies. It does not remove the hypotheses in the statement or the obligations carried by its structures.

2 Exact and approximate gluing

2.1 The sheaf used by the development

Let X be a topological space with the measurable and Borel structure required by the formal statement. Let F_0 assign to an open subset its finite spatial measures, with restriction maps. The Lean definition `probabilityPresheaf` denotes the sheafification F of F_0 . Despite that identifier, the initial local data are unnormalised finite measures; mass one is not part of this definition.

Claim 2.1. *F is a sheaf.*

Proof. Sheafification returns a sheaf together with its sheaf-property proof. Projecting that proof establishes the claim for F . $\square$

Extracted Lean statement: Appendix A.1.

Scope. This result uses the sheafification construction. It supplies no physical mechanism for producing compatible local data.

2.2 Compatible local sections have a unique extension

Claim 2.2. *Let F be any sheaf of types on X , and let $(U_i)_{i \in I}$ be an open cover of X . Suppose $s_i \in F(U_i)$ and*

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j} \quad (i, j \in I).$$

Then exactly one $g \in F(X)$ satisfies $g|_{U_i} = s_i$ for every i .

Proof. The unique-gluing form of the sheaf condition gives a section on $\bigcup_i U_i$, with the specified restrictions, unique among sections there. The cover equality identifies that union with X . Transport the section along this equality. Functoriality of the restriction maps shows that its restriction to each U_i is s_i .

Conversely, transport any competing section on X back to the union. Its restrictions are again the s_i , so the uniqueness supplied by the sheaf condition identifies it with the first section. Transporting forward gives the required equality on X . In the Lean proof, equality of the relevant inclusion maps uses the fact that morphisms between opens form a subsingleton. $\square$

Extracted Lean statement: Appendix A.2.

Claim 2.3. *The same conclusion holds for the development's sheafified spatial measures.*

Proof. Apply the preceding claim to $F = \text{probabilityPresheaf}$ and use the sheaf property established above. The compatible family and cover equality are exactly the hypotheses of this application. $\square$

Extracted Lean statement: Appendix A.3.

Scope. This proof is a direct term applying a previous theorem. It has no tactic history; the extracted statement and proof dependencies still cover it.

2.3 What equilibrium contributes

A thermodynamic cover supplies local sections s_i , phases θ_i , and a finite, symmetric, strictly positive coupling matrix. Its structure includes two substantive conditions: the phase configuration globally minimises the reduced Kuramoto potential, and

$$\cos(\theta_i - \theta_j) = 1 \implies s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

The second condition is an obligation on the supplied local data.

Claim 2.4. *A thermodynamic cover determines a unique global section with its prescribed local restrictions.*

Proof. The theorem characterising global minima of the reduced Kuramoto potential implies phase locking for the cover: $\cos(\theta_i - \theta_j) = 1$ for every pair. The overlap-agreement field then makes the local sections compatible. Apply the spatial-measure gluing theorem to the cover and that compatible family. $\square$

Extracted Lean statement: Appendix A.8.

Scope. This argument consumes the supplied equilibrium and overlap conditions. The existence of a physical system satisfying them is a separate matter.

2.4 When the section represents an actual finite measure

Let $(U_i)_{i \in I}$ be a finite measurable cover of X and let μ_i be finite measures, extended by zero outside their patches if necessary. Assume

$$\mu_i|_{U_i \cap U_j} = \mu_j|_{U_i \cap U_j} \quad (i, j \in I).$$

These are equalities of actual measures, before sheafification.

Claim 2.5. *There is a unique finite measure ν whose restriction to each U_i is $\mu_i|_{U_i}$.*

Proof. Order the finite index set and put $D_i = U_i \setminus \bigcup_{j < i} U_j$. The D_i are measurable, disjoint, cover X , and satisfy $D_i \subseteq U_i$. Define $\nu = \sum_i \mu_i|_{D_i}$, a finite sum of finite measures. On $U_k \cap D_i$, overlap agreement identifies μ_i with μ_k . Thus $\nu|_{U_k} = \sum_i \mu_k|_{U_k \cap D_i} = \mu_k|_{U_k}$. Two measures with these restrictions agree on the covering pieces, hence are equal. $\square$

Extracted Lean statement: Appendix A.122.

Claim 2.6. *If a thermodynamic cover's local sections are the sheaf images of such compatible finite measures, its glued section is the image of ν .*

Proof. The sheafification map preserves restrictions. The equality of actual measure restrictions therefore makes the image of ν restrict to every supplied local section. Uniqueness of the sheaf gluing identifies this image with `ThermodynamicCover.invariantMeasure`. $\square$

Extracted Lean statement: Appendix A.24.

Scope. The cover already has a finite patch index. Compatible finite-measure representatives are explicit additional data; no general inverse to sheafification is assumed. Local finite measures can always be extended by zero to the ambient space and recovered on their patches (measureOnOpen_map_subtype). No mass-one normalization is imposed.

On the real line, the patches $(-\infty, 1)$ and $(-1, \infty)$ carry $\delta_{-2} + \delta_0$ and $\delta_0 + \delta_2$, respectively. They agree on the overlap, and their unique extension is $\delta_{-2} + \delta_0 + \delta_2$, with total mass three.

Extracted Lean statement: Appendix A.124.

Extracted Lean statement: Appendix A.125.

By contrast, singleton patches at every natural number, each requiring mass one, have no finite-measure extension: any such measure would equal counting measure, whose total mass is infinite.

Extracted Lean statement: Appendix A.123.

2.5 Approximate agreement gives a controlled selection

Let the site set and patch index set be finite. Let $s_i(x) \geq 0$ be local mass profiles represented as functions on the ambient sites. Let $w_i(x) \geq 0$ satisfy $w_i(x) = 0$ outside U_i and $\sum_i w_i(x) = 1$ at every site. Assume that for $x \in U_i \cap U_j$, $|s_i(x) - s_j(x)| \leq \varepsilon$, where $\varepsilon \geq 0$.

Claim 2.7. *The selected profile $g(x) = \sum_j w_j(x)s_j(x)$ satisfies $\sup_{x \in U_i} |g(x) - s_i(x)| \leq \varepsilon$ for every patch i .*

Proof. Fix $x \in U_i$. Every nonzero weight $w_j(x)$ has $x \in U_j$, so its profile is within ε of $s_i(x)$. Normalisation and the triangle inequality therefore give

$$|g(x) - s_i(x)| = \left| \sum_j w_j(x)(s_j(x) - s_i(x)) \right| \leq \sum_j w_j(x)\varepsilon = \varepsilon.$$

The terms with zero weight contribute zero. The uniform metric on the finite patch converts these pointwise bounds into the displayed profile-distance bound. Lean isolates the weighted inequality in its helper `weighted_dist_le` and then applies it at each site. $\square$

Extracted Lean statement: Appendix A.15.

Scope. The profiles need not have total mass one. The result provides an approximation in the specified uniform mass metric, and assumes the overlap bounds and the subordinate normalised weights.

3 The stationary critical exponent

Write

$$E(a) = \frac{\int_0^{2\pi} \sin^2 \theta e^{a \cos \theta} d\theta}{\int_0^{2\pi} e^{a \cos \theta} d\theta}.$$

This is `vonMisesSRatio`. Its denominator is strictly positive.

3.1 The second-order expansion

Claim 3.1. As $a \rightarrow 0$,

$$E(a) = \frac{1}{2} - \frac{a^2}{16} + o(a^2).$$

Proof. The previously proved differentiability results give $E \in C^2$, $E(0) = 1/2$, $E'(0) = 0$, and $E''(0) = -1/8$. Taylor's theorem with a little-o remainder yields $E(a) = E(0) + E'(0)a + E''(0)a^2/2 + o(a^2)$. Substituting those values gives the claimed polynomial. The Lean proof applies the general Taylor theorem, identifies its second iterated derivative using the derivative lemmas, and simplifies its Taylor polynomial. $\square$

Extracted Lean statement: Appendix A.41.

Scope. The asserted remainder is $o(a^2)$. This statement supplies no fourth-order remainder estimate.

3.2 An arbitrary family approaching threshold

Claim 3.2. Let $D > 0$, let ℓ be a filter on an index type, and let K_x, r_x be real-valued functions on that type. Suppose $K_x \rightarrow 2D$, $r_x \rightarrow 0$ along ℓ , and eventually

$$K_x > 2D, \quad r_x > 0, \quad r_x = \text{selfConsistency}(K_x, D, r_x).$$

Then $r_x^2/(K_x - 2D) \rightarrow 1/D$ along ℓ .

Proof. Set $a_x = K_x r_x / D$ and write

$$E(a) = \frac{1}{2} - \frac{a^2}{16} + a^2 q(a).$$

The previous expansion gives $q(a) \rightarrow 0$ as $a \rightarrow 0$, taking the quotient with Lean's totalised division at zero. The assumed limits imply $a_x \rightarrow 0$, hence $q(a_x) \rightarrow 0$.

For an index satisfying the eventual hypotheses, positivity gives $a_x \neq 0$ and $K_x - 2D \neq 0$. The previously proved reformulation of the self-consistency equation gives $E(a_x) = D/K_x$. Substituting the expansion and rearranging yields

$$K_x - 2D = K_x a_x^2 (1/8 - 2q(a_x)), \quad \frac{r_x^2}{K_x - 2D} = \frac{D^2}{K_x^3 (1/8 - 2q(a_x))}.$$

The denominator in the last expression tends to $(2D)^3/8 = D^3 > 0$. The quotient therefore tends to $D^2/D^3 = 1/D$. The equality of the quotients holds eventually, which transfers this limit to the desired expression. $\square$

Extracted Lean statement: Appendix A.42.

Scope. Convergence to zero and eventual stationarity are hypotheses. This general statement allows the bottom filter, where its eventual and convergence conditions are vacuous. A concrete branch supplies the substantive application below. The claim establishes stationary asymptotics, with no time evolution or stability conclusion.

3.3 A branch that satisfies the hypotheses

For $D > 0$ and $K > 2D$, define $r_*(D, K)$ by choosing the coherent solution from the theorem that proves existence and uniqueness in $(0, 1]$. The Lean function `coherentBranch` uses this choice in that regime and returns zero outside it.

Claim 3.3. *If $D > 0$ and $K > 2D$, then $0 < r_*(D, K) \leq 1$ and*

$$r_*(D, K) = \text{selfConsistency}(K, D, r_*(D, K)).$$

Proof. The two inequalities on D, K select the positive branch of the definition. The specification of the chosen witness is exactly positivity, the bound by one, and the fixed-point equation supplied by the existence theorem. $\square$

Extracted Lean statement: Appendix A.43.

Claim 3.4. *For fixed $D > 0$, as K decreases to $2D$ through $K > 2D$,*

$$\frac{r_*(D, K)^2}{K - 2D} \longrightarrow \frac{1}{D}.$$

Proof. Apply the family theorem to the right-hand neighbourhood filter of $2D$, with the identity function for K . The previously proved `coherentBranch_tendsto_zero` supplies the limit of r_* at the threshold. Membership in the right-hand neighbourhood gives $K > 2D$ eventually, and the branch specification supplies positivity and stationarity there. These are all the hypotheses of the family theorem. $\square$

Extracted Lean statement: Appendix A.45.

Scope. The approach filter on the real line is nontrivial, and the branch provides actual coherent solutions. This removes the vacuity issue in this application. The conclusion still concerns stationary solutions.

4 What the conditional chain composes

The chain uses a common topological and measured substrate, its sheaf of global sections, a supplied complete metric on those sections, a neural field, a thermodynamic cover T , and a reflexive boundary with prediction map P . The precise type and instance parameters are printed in the appendix.

4.1 The common downstream argument

Suppose a sequence of energies coarse-grains to L , the elapsed parameter τ is positive, and the following named implications hold:

- *E56:* coarse-graining implies $D > 0$, $K \geq 0$, $L = K$, and an identification of the supplied field with the parameters K, D .
- *E67:* those scalar conditions imply $K > 2D$.
- *E78:* a coherent order parameter implies that the supplied cover is reached by the specified relaxation at a coupling at least K .
- *E89:* that reachability supplies a section s which restricts to the local data of T , is fixed by P , and comes with the stated Lipschitz bound on P at $\text{resonanceRate}(K, D, \tau)$.

Claim 4.1. *There is a global section compatible with a thermodynamic cover which is the unique fixed point of the prediction map.*

Proof. Apply *E56* to the coarse-graining hypothesis and *E67* to its scalar conclusion. The supercritical existence theorem supplies a coherent order parameter. Apply *E78* and then *E89* to obtain the section s , its local restrictions, the Lipschitz bound, and $P(s) = s$.

The section s makes the space of global sections nonempty. The theorem `self_of_coherent_order_parameter`, using completeness, the Lipschitz bound, coherence, and $\tau > 0$, applies the contraction argument to obtain a unique fixed point p . Since s is fixed, uniqueness gives $s = p$. Every fixed point t equals p and hence s . Pair s with the supplied cover T and its known restriction equations. This is exactly the predicate `UnifiedSelf`. $\square$

Extracted Lean statement: Appendix A.69.

Scope. The identity between the glued section and a fixed section is supplied in *E89*; the contraction argument proves uniqueness. The concrete-field conjunct of *E56* is retained as an input constraint but is not used by this proof's downstream fixed-point argument. The source also establishes a local `Unity` fact which the concluding construction does not reference. Its constants can still occur in the unnormalised proof term; a syntactic dependency list alone therefore does not identify minimal premises.

4.2 The passive composition

The passive chain additionally supplies a finite nonempty register, its statistical-mechanics instance and update, a field and vacuum set, and measurable carrier types for predictive information. Its remaining named implications are:

- *E12*: the capacity bound implies that the field leaves the vacuum set.
- *E23*: leaving that set implies the register update is not surjective.
- *E34*: dissipation supplies a predictive-dissipation structure whose thermal scale and dissipated work match the register's temperature and heat, and whose nonpredictive information is strictly positive.
- *E45*: the predictive bound implies the chosen coarse-graining statement.

Claim 4.2. *Under *E12* through *E89* and the stated structural hypotheses, the passive chain concludes `UnifiedSelf` for the supplied reflexive boundary.*

Proof. Finiteness gives the capacity bound. Apply *E12* to obtain vacuum departure, then *E23* to obtain a nonsurjective update. The proved dissipation theorem, using the supplied statistical-mechanics structure, gives positive heat. Apply *E34* and retain its predictive-dissipation witness R . The bound associated with R gives the predictive node. Apply *E45* to obtain coarse-graining and then invoke the downstream claim with *E56* through *E89*. $\square$

Extracted Lean statement: Appendix A.70.

Scope. The eight named implications remain hypotheses. Conditions inside the statistical-mechanics, thermodynamic-cover, and other structures are additional obligations. The equalities and positivity in *E34* constrain which witness is admitted, while the passage through *E45* uses its predictive structure. The conclusion has no predicate asserting experience or biological implementation.

5 That the chain is not vacuous, and what it still does not say

Both chain theorems are conditional: each accepts eight named implications and concludes `UnifiedSelf`. A conditional theorem whose hypotheses nothing satisfies proves nothing about anything, so the development supplies witnesses discharging all eight at once, and --- separately --- counterexamples showing that the same hypotheses cannot be met by substituting an arbitrary target.

5.1 All eight arrows in one composition

Claim 5.1. *There are a double well, an absorbing register, an erasing update, a refining mesh, and a field and cover over the type `Cortex` that satisfy `E12` through `E89` simultaneously; hence `UnifiedSelf` holds for the supplied reflexive boundary.*

Proof. Apply the passive chain to those components. The first four arrows are supplied by their witnesses; `E56` is discharged by the numerals $D = 1$, $K = 3$, $L = 3$ together with the identification of the supplied field, `E67` by evaluating the critical coupling and comparing $3 > 2$, `E78` by the cover's reachability at coupling three, and `E89` by the constructed section, its restriction equations, the Lipschitz bound and the fixed-point equation. $\square$

Extracted Lean statement: Appendix A.95.

Scope. The components are separate toy constructions with different state spaces. The field, cover and reflexive boundary share the three-site substrate. Their simultaneous satisfaction establishes that the hypotheses are consistent, and identifies no physical mechanism common to a double well, a register, a mesh and a cortical cover. In particular nothing here interprets `Cortex` as `cortex`.

5.2 Four active branches, not one

The active chain replaces the passive predictive node with a heat-budget node. The development discharges its hypotheses with four arrangements: with a noisy actuator, with that actuator held to a named register budget, with a process-dependent actuated field, and with a kernel built from mode occupancies on a product space.

Extracted Lean statement: Appendix A.106.

Extracted Lean statement: Appendix A.109.

Extracted Lean statement: Appendix A.114.

Extracted Lean statement: Appendix A.120.

Scope. The register-budget arrangement changes `E34Active`: its heat allocation is derived from the register's ledger. The actuated and microscopic arrangements change `E45Active`, the arrow feeding coarse-graining, using process-dependent spatial energies. All four retain the same downstream field, cover and boundary. They demonstrate several ways to satisfy the hypotheses; they do not identify one common physical mechanism.

5.3 The obligations are not free

Two kinds of rejection fence the active node. The first concerns the budget.

Claim 5.2. *The actuator does not satisfy `ActiveBound` at budget 0.*

Proof. `ActiveBound` requires the process's cost to lie below the budget. The actuator's cost is positive, and a positive real is not at most zero. $\square$

Extracted Lean statement: Appendix A.103.

The second concerns the limit. Each convergence arrow says that, *given* the bound, a named energy sequence tends to a named target. Since the bound genuinely holds for these processes, the arrow can be applied, and the target is then pinned by uniqueness of limits.

Claim 5.3. *The active convergence arrow fails for the constant-zero energy sequence at target 3; for the actuated sequence at target 0; and for the microscopic sequence at target 0. The passive arrow likewise fails for the remembered sequence at target 0.*

Proof. In each case supply the arrow with the bound that does hold, obtaining convergence of the sequence to the stated target. The same sequence is already known to converge --- the constant sequence to 0, the others to their computed spatial limits. Uniqueness of limits in $\mathbb{R}$ equates the two values, and the resulting numerical identity is false. $\square$

Extracted Lean statement: Appendix A.104.

Extracted Lean statement: Appendix A.113.

Extracted Lean statement: Appendix A.116.

Extracted Lean statement: Appendix A.119.

Scope. These are rejections rather than vacuous implications precisely because the premise is inhabited in each case. They show that the convergence arrows constrain which energy sequence and which limit may be supplied; they do not show that any particular sequence is the physically correct one. Choosing it remains a modelling obligation outside the theorem.

5.4 The passive edge does not pick out its hardware

Claim 5.4. *Two different declared systems over the same joint law --- one retaining its memory, one scrambling it --- discharge the identical passive convergence edge.*

Proof. Each supplies the joint law that the edge's construction requires, and the convergence itself is the same spatial fact about the shared energy sequence. Neither proof consults the system's efficiency. $\square$

Extracted Lean statement: Appendix A.117.

Extracted Lean statement: Appendix A.118.

Scope. The edge is therefore insensitive to how well the hardware preserves what it stores. No configuration law, partition or efficiency is selected by discharging it, which bounds how much the passage through this node can be read as a claim about physical implementation.

6 The vertex-supported obstruction and the cell-pair construction

The scalar mesh-refinement theorem does not reconstruct a continuum kernel. Its vertex-supported candidate contributes zero to the continuum coupling functional on an atomless substrate. This obstruction concerns that particular representation. The development also provides a cell-pair construction which converges to a specified continuous kernel.

6.1 The target structure constrains nothing

A `ContinuousNeuralField` over a substrate M carries a drift $\omega : M \rightarrow \mathbb{R}$, a kernel $K : M \times M \rightarrow \mathbb{R}$, and a timescale $\tau \in \mathbb{R}$, with no conditions relating them.

Claim 6.1. *For any ω , K and τ whatsoever, a `ContinuousNeuralField` over M exists.*

Proof. The three data are exactly the structure's fields, so the anonymous constructor supplies the required inhabitant. $\square$

Extracted Lean statement: Appendix A.96.

Scope. A constructor taking a mesh to such a field would therefore typecheck with a kernel unrelated to that mesh, and its proof would be a definitional unfolding. The statement records that the edge would carry no information, not that a limit fails to converge.

6.2 The candidate kernel does carry the weights

Write emb for a triangulation's vertex embedding into M and w for its edge weights. The induced kernel places each weight at its embedded pair,

$$\text{vertexKernel}(x, y) = \sum_u \sum_v [x = \text{emb } u \wedge y = \text{emb } v] w(u, v),$$

and vanishes off the image of the vertex set.

Claim 6.2. *The kernel is sited on the finite set $\text{emb}[V]$: if $x \notin \text{emb}[V]$ then $\text{vertexKernel}(x, y) = 0$ for every y .*

Proof. Fix such an x and any y . Every summand carries the condition $x = \text{emb } u$; were it met for some u , x would lie in the image. So each summand is zero, and both sums vanish. $\square$

Extracted Lean statement: Appendix A.97.

Claim 6.3. *If emb is injective, then $\text{vertexKernel}(\text{emb } u, \text{emb } v) = w(u, v)$.*

Proof. Injectivity makes $\text{emb } u = \text{emb } b$ fail for $b \neq u$, so the outer sum reduces to its u th term, and the inner sum to its v th by the same argument. That term is $w(u, v)$. $\square$

Extracted Lean statement: Appendix A.98.

Scope. This is what stops the next result from being a statement about a kernel that was zero from the outset. At embedded pairs the kernel takes precisely the value the triangulation assigns.

6.3 The no-go

The field results are stated through the coupling functional

$$\text{fieldCorrelation}(\mu, \theta, K) = \int_M \left(\int_M K(x, y) \cos(\theta(y) - \theta(x)) d\mu(y) \right) d\mu(x).$$

Claim 6.4. *On a substrate whose measure has no atoms, $\text{fieldCorrelation}(\mu, \theta, \text{vertexKernel}) = 0$ --- for every weight assignment and every phase field.*

Proof. By the siting claim the inner integral vanishes for each x outside the finite set $\text{emb}[V]$. A finite set is a finite union of singletons, and no singleton carries mass, so that set is null. The outer integral of a function supported in a null set is zero. $\square$

Extracted Lean statement: Appendix A.99.

Scope. The weights are unconstrained and the phase field is arbitrary, so no choice of triangulation data evades the conclusion. The obstruction is that the discrete data occupies a null set, not that the weights are small.

6.4 What this locates

The coarse-graining theorem converges a discrete coupling *energy* --- one real number per triangulation --- to an integral over the substrate, and the edge-weight sum reaches the same scalar. Both integrate the pair structure away, while a continuum kernel is a function of two points. Spreading the weights over cells rather than vertices would require a product-structured decomposition of $M \times M$; the triangulation's edge regions are subsets of M alone.

6.5 The positive construction and its input

KernelMesh supplies a finite measurable partition and one sample per cell. It extends each sampled pair value over the whole cell pair. KernelMesh.kernel_tendsto proves pointwise convergence to a supplied continuous kernel when the samples approach their points; KernelMesh.energy_tendsto passes the limit through the integral under compactness, second countability and finite measure. The microscopic actuator provides one such kernel from declared mode profiles and occupancies. These results are in Phase2_KernelMesh.lean and Phase3_LocalActuator.lean. The cell-pair integral identity and the two convergence results are included in the extracted appendix.

Extracted Lean statement: Appendix A.127.

Extracted Lean statement: Appendix A.128.

Extracted Lean statement: Appendix A.129.

Scope. The vertex-supported no-go does not obstruct this cell-pair construction. Its limiting kernel is specified by the hardware model, rather than inferred from a scalar energy alone. The chain receives a concrete field through E56, whose kernel conditions constrain the admissible input. The downstream fixed-point proof uses only the scalar part of E56; the field conditions are consumed separately by em_field_exhibits_phase_transition. Identifying that field with cortex remains a physical modelling obligation.

7 A concrete nonconstant family and its trace

On the three-site type with sites left, mid, and right, the development defines the nonnegative mass profile

$$d_t = (\max(1 + \cos t, 0), 1, \max(2 + \cos t, 0)).$$

The use of max expresses Real.toNNReal; these are nonnegative spatial masses, without a normalisation requirement.

Claim 7.1. $d_0 \neq d_\pi$.

Proof. Suppose the profiles were equal. Evaluating at the left site would equate $\max(1 + \cos 0, 0) = 2$ and $\max(1 + \cos \pi, 0) = 0$, a contradiction. $\square$

Extracted Lean statement: Appendix A.121.

Scope. This witnesses nonconstant dependence on phase for this particular family. It does not assert that every thermodynamic cover has such dependence, or that this family models a biological substrate.

7.1 Before and after the actual tactics

The following excerpt is generated from the saved elaboration data for this proof. It selects the contradiction assumption, the evaluation at the left site, and the simplification that closes the goal. The complete nested records, including wrappers sharing source spans, remain in `data/witness.json`.

```
SOURCE LINE 101: intro h
BEFORE
⊢ richDensity 0 ≠ richDensity Real.pi
AFTER
h : richDensity 0 = richDensity Real.pi
⊢ False

SOURCE LINE 102: have h1 := congrFun h Site.left
BEFORE
h : richDensity 0 = richDensity Real.pi
⊢ False
AFTER
h : richDensity 0 = richDensity Real.pi
h1 : richDensity 0 Site.left = richDensity Real.pi Site.left
⊢ False

SOURCE LINE 103: simp [richDensity] at h1
BEFORE
h : richDensity 0 = richDensity Real.pi
h1 : richDensity 0 Site.left = richDensity Real.pi Site.left
⊢ False
AFTER
No goals remain.
```

A Extracted statements and direct project dependencies

The following text is generated from elaborated Lean expressions. Universe parameter names, complete dependency lists including Mathlib, source excerpts, and goal states are retained in the JSON snapshots. Pretty-printing uses Lean's mathematical notation and can suppress implicit arguments at applications; the quantified hypotheses of the declaration remain in the displayed type. Dependency lists here show direct project constants in the proof term. They are not a minimal set of assumptions or a transitive explanation of every lemma.

A.1 sheaf-property

PhysicsOfConsciousness.probability_is_sheaf

Source, line 22; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X],
  TopCat.Presheaf.IsSheaf (PhysicsOfConsciousness.probabilityPresheaf X)
```

Direct project constants in the proof term:

PhysicsOfConsciousness.probabilityPresheaf_pre

A.2 sheaf-gluing

PhysicsOfConsciousness.sheaf_glue_unique

Source, line 55; 494 nested tactic records.

```
∀ {X : TopCat} {F : (TopologicalSpace.OPens ↑X)op ⇒ Type u},
  TopCat.Presheaf.IsSheaf F →
  ∀ {I : Type u} (cover : I → TopologicalSpace.OPens ↑X),
  iSup cover = τ →
  ∀ (s : (i : I) → F.obj (Opposite.op (cover i))),
  (∀ (i j : I),
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) (s i) =
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) (s j)) →
  ∃! g,
  ∀ (i : I),
    (CategoryTheory.ConcreteCategory.hom (F.map (CategoryTheory.homOfLE ↗).op)) g = s i
```

Direct project constants in the proof term:

PhysicsOfConsciousness.sheaf_glue_unique.match_1

A.3 probability-gluing

PhysicsOfConsciousness.probability_glue_unique

Source, line 111; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X] {I : Type u}
  (cover : I → TopologicalSpace.OPens ↑X),
  iSup cover = τ →
  ∀ (s : (i : I) → (PhysicsOfConsciousness.probabilityPresheaf X).obj (Opposite.op (cover i))),
  (∀ (i j : I),
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↗).op))
      (s i) =
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↗).op))
      (s j)) →
  ∃! g,
  ∀ (i : I),
```

```

(CategoryTheory.ConcreteCategory.hom
  ((PhysicsOfConsciousness.probabilityPresheaf X).map
    (CategoryTheory.homOfLE  $\mapsto$ ).op))
  g =
  s i

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.probability_is_sheaf
PhysicsOfConsciousness.sheaf_glue_unique

```

A.4 a-symm

PhysicsOfConsciousness.ThermodynamicCover.A_symm

[Source, line 252](#); 0 nested tactic records.

```

∀ {X : TopCat} {inst : MeasurableSpace  $\mapsto$  X} {inst_1 : BorelSpace  $\mapsto$  X}
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold  $\mapsto$  X}
  [self : PhysicsOfConsciousness.ThermodynamicCover X]
  (i j : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
  PhysicsOfConsciousness.ThermodynamicCover.A i j = PhysicsOfConsciousness.ThermodynamicCover.A j i

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.TriangulatedManifold

```

A.5 a-pos

PhysicsOfConsciousness.ThermodynamicCover.A_pos

[Source, line 253](#); 0 nested tactic records.

```

∀ {X : TopCat} {inst : MeasurableSpace  $\mapsto$  X} {inst_1 : BorelSpace  $\mapsto$  X}
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold  $\mapsto$  X}
  [self : PhysicsOfConsciousness.ThermodynamicCover X]
  (i j : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
  PhysicsOfConsciousness.ThermodynamicCover.A i j > 0

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.TriangulatedManifold

```

A.6 thermodynamic-equilibrium

PhysicsOfConsciousness.ThermodynamicCover.thermodynamic_equilibrium

[Source, line 254](#); 0 nested tactic records.

```

∀ {X : TopCat} {inst : MeasurableSpace  $\mapsto$  X} {inst_1 : BorelSpace  $\mapsto$  X}
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold  $\mapsto$  X}
  [self : PhysicsOfConsciousness.ThermodynamicCover X]
  (theta : PhysicsOfConsciousness.LocalSectionSynchronization.I X  $\rightarrow$   $\mathbb{R}$ ),
  PhysicsOfConsciousness.kuramoto_potential_dynamic
    { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm :=  $\dots$  }
    PhysicsOfConsciousness.LocalSectionSynchronization.phase  $\leq$ 
    PhysicsOfConsciousness.kuramoto_potential_dynamic
    { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm :=  $\dots$  } theta

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.TriangulatedManifold

```

A.7 phase-locked

PhysicsOfConsciousness.ThermodynamicCover.phase_locked

Source, line 270; 29 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  (T : PhysicsOfConsciousness.ThermodynamicCover X),
  PhysicsOfConsciousness.is_phase_locked PhysicsOfConsciousness.LocalSectionSynchronization.phase
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.KuramotoSystem.mk
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.phase
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.A
PhysicsOfConsciousness.ThermodynamicCover.A_pos
PhysicsOfConsciousness.ThermodynamicCover.A_symm
PhysicsOfConsciousness.ThermodynamicCover.I_decidable
PhysicsOfConsciousness.ThermodynamicCover.I_fintype
PhysicsOfConsciousness.ThermodynamicCover.thermodynamic_equilibrium
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.potential_min_implies_phase_locked
```

A.8 thermodynamic-gluings

PhysicsOfConsciousness.global_section_from_thermodynamics

Source, line 303; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [T : PhysicsOfConsciousness.ThermodynamicCover X],
  ∃! s,
    ∀ (i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
      (CategoryTheory.ConcreteCategory.hom
        ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↪).op))
        s =
          PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.is_cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.phase_locked
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.overlap_agreement
PhysicsOfConsciousness.probability_glue_unique
```

A.9 sync-to-section-eq

PhysicsOfConsciousness.ThermodynamicCover.sync_to_section_eq

Source, line 348; 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  (T : PhysicsOfConsciousness.ThermodynamicCover X)
```

```

(i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i =
  (CategoryTheory.ConcreteCategory.hom
    ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE  $\mapsto$ ).op))
    T.invariantMeasure

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.global_section_from_thermodynamics
PhysicsOfConsciousness.probabilityPresheaf

```

A.10 invariantmeasure-unique

PhysicsOfConsciousness.ThermodynamicCover.invariantMeasure_unique

[Source, line 356](#); 0 nested tactic records.

```

∀ {X : TopCat} [inst : MeasurableSpace  $\mapsto$  X] [inst_1 : BorelSpace  $\mapsto$  X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold  $\mapsto$  X]
  (T : PhysicsOfConsciousness.ThermodynamicCover X) (s : PhysicsOfConsciousness.GlobalSection),
  (∀ (i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE  $\mapsto$ ).op))
      s =
        PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i) →
    s = T.invariantMeasure

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.global_section_from_thermodynamics
PhysicsOfConsciousness.probabilityPresheaf

```

A.11 profiledist-le-iff

PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff

[Source, line 490](#); 17 nested tactic records.

```

∀ {α : Type u_1} [inst : Fintype α] {U : Set α} {f g : α → NNReal} {ε : ℝ},
  0 ≤ ε →
  (PhysicsOfConsciousness.ApproximateGluing.profileDist U f g ≤ ε ↔ ∀ x ∈ U, dist (f x) (g x) ≤ ε)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ApproximateGluing.profileDist._proof_1

```

A.12 compatible-pointwise

PhysicsOfConsciousness.ApproximateGluing.compatible_pointwise
[Source, line 509](#); 24 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] {U : ι → Set α} {s : ι → α → NNReal}
  {ε : NNReal},
  PhysicsOfConsciousness.ApproximateGluing.Compatible U s ε →
  PhysicsOfConsciousness.LocalContent.Compatible U (fun i x => ↑(s i x)) ↑ε
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.Compatible
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
```

A.13 covers

PhysicsOfConsciousness.ApproximateGluing.IsPartition.covers
[Source, line 530](#); 62 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype ι] {U : ι → Set α} {w : ι → α → NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w → ∀ (x : α), ∃ i, x ∈ U i
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.IsPartition
```

A.14 select-eq-of-eqon

PhysicsOfConsciousness.ApproximateGluing.select_eq_of_eqon
[Source, line 541](#); 82 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype ι] {U : ι → Set α} {w s t : ι → α → NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w →
  (∀ (i : ι), Set.EqOn (s i) (t i) (U i)) →
  PhysicsOfConsciousness.ApproximateGluing.select w s =
  PhysicsOfConsciousness.ApproximateGluing.select w t
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.IsPartition
PhysicsOfConsciousness.ApproximateGluing.select
```

A.15 approximate-gluing

PhysicsOfConsciousness.ApproximateGluing.select_close
[Source, line 576](#); 47 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] [inst_1 : Fintype ι] {U : ι → Set α}
  {w s : ι → α → NNReal} {ε : NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w →
  PhysicsOfConsciousness.ApproximateGluing.Compatible U s ε →
  ∀ (i : ι),
    PhysicsOfConsciousness.ApproximateGluing.profileDist (U i)
      (PhysicsOfConsciousness.ApproximateGluing.select w s) (s i) ≤
    ↑ε
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.Compatible
PhysicsOfConsciousness.ApproximateGluing.IsPartition
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
PhysicsOfConsciousness.ApproximateGluing.select
```

A.16 select-dist-le

PhysicsOfConsciousness.ApproximateGluing.select_dist_le
[Source, line 593](#); 102 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] [inst_1 : Fintype ι] {U : ι → Set α}
  {w v s : ι → α → NNReal} {ε : NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w →
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U v →
  PhysicsOfConsciousness.ApproximateGluing.Compatible U s ε →
  dist (PhysicsOfConsciousness.ApproximateGluing.select w s)
    (PhysicsOfConsciousness.ApproximateGluing.select v s) ≤
    ↑ε
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.Compatible
PhysicsOfConsciousness.ApproximateGluing.IsPartition
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
PhysicsOfConsciousness.ApproximateGluing.select
PhysicsOfConsciousness.ApproximateGluing.select_close
```

A.17 approximate-diameter-le

PhysicsOfConsciousness.ApproximateGluing.approximate_diameter_le
[Source, line 611](#); 45 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] {U : ι → Set α} {s : ι → α → NNReal}
  {g h : α → NNReal} {δ : NNReal},
  (∀ (x : α), ∃ i, x ∈ U i) →
  (∀ (i : ι), PhysicsOfConsciousness.ApproximateGluing.profileDist (U i) g (s i) ≤ ↑δ) →
  (∀ (i : ι), PhysicsOfConsciousness.ApproximateGluing.profileDist (U i) h (s i) ≤ ↑δ) →
  dist g h ≤ 2 * ↑δ
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
```

A.18 select-glue-unique

PhysicsOfConsciousness.ApproximateGluing.select_glue_unique
[Source, line 629](#); 69 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] [inst_1 : Fintype ι] {U : ι → Set α}
  {w s : ι → α → NNReal},
  PhysicsOfConsciousness.ApproximateGluing.IsPartition U w →
  PhysicsOfConsciousness.ApproximateGluing.Compatible U s 0 →
  ∃! g, ∀ (i : ι), Set.EqOn g (s i) (U i)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluing.Compatible
PhysicsOfConsciousness.ApproximateGluing.IsPartition
PhysicsOfConsciousness.ApproximateGluing.IsPartition.covers
PhysicsOfConsciousness.ApproximateGluing.approximate_diameter_le
PhysicsOfConsciousness.ApproximateGluing.profileDist
PhysicsOfConsciousness.ApproximateGluing.profileDist_le_iff
PhysicsOfConsciousness.ApproximateGluing.select
PhysicsOfConsciousness.ApproximateGluing.select_close
```

A.19 feature-dist-le

PhysicsOfConsciousness.ApproximateGluings.feature_dist_le

[Source, line 649](#); 0 nested tactic records.

```
∀ {α : Type u_1} {ι : Type u_2} [inst : Fintype α] [inst_1 : Fintype ι] {U : ι → Set α}
  {w v s : ι → α → NNReal} {ε : NNReal} {Y : Type u_3} [inst_2 : PseudoMetricSpace Y] {L : NNReal}
  {F : (α → NNReal) → Y},
  LipschitzWith L F →
  PhysicsOfConsciousness.ApproximateGluings.IsPartition U w →
  PhysicsOfConsciousness.ApproximateGluings.IsPartition U v →
  PhysicsOfConsciousness.ApproximateGluings.Compatible U s ε →
  dist (F (PhysicsOfConsciousness.ApproximateGluings.select w s))
    (F (PhysicsOfConsciousness.ApproximateGluings.select v s)) ≤
    ↑L * ↑ε
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ApproximateGluings.Compatible
PhysicsOfConsciousness.ApproximateGluings.IsPartition
PhysicsOfConsciousness.ApproximateGluings.select
PhysicsOfConsciousness.ApproximateGluings.select_dist_le
```

A.20 measure-zero-extension

PhysicsOfConsciousness.measureOnOpen_map_subtype

[Source, line 377](#); 6 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [BorelSpace ↑X] (U : TopologicalSpace.opens ↑X)
  (μ : MeasureTheory.FiniteMeasure ↑U),
  PhysicsOfConsciousness.measureOnOpen (μ.map Subtype.val) U = μ
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.measureOnOpen
```

A.21 measure-restriction

PhysicsOfConsciousness.measureOnOpen_restrict

[Source, line 389](#); 7 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  (μ : MeasureTheory.FiniteMeasure ↑X) {U V : TopologicalSpace.opens ↑X} (h : V ≤ U),
  (CategoryTheory.ConcreteCategory.hom
    ((PhysicsOfConsciousness.probabilityPresheaf_pre X).map (CategoryTheory.homOfLE h).op))
    (PhysicsOfConsciousness.measureOnOpen μ U) =
    PhysicsOfConsciousness.measureOnOpen μ V
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.inc_is_measurable_embedding
PhysicsOfConsciousness.measureOnOpen
PhysicsOfConsciousness.probabilityPresheaf_pre
```

A.22 measure-section-restriction

PhysicsOfConsciousness.globalSectionOfMeasure_restrict

[Source, line 399](#); 18 nested tactic records.

```

∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  (μ : MeasureTheory.FiniteMeasure ↑X) (U : TopologicalSpace.Opens ↑X),
  (CategoryTheory.ConcreteCategory.hom
    ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↪).op))
    (PhysicsOfConsciousness.globalSectionOfMeasure μ) =
  (CategoryTheory.ConcreteCategory.hom
    ((TopCat.Presheaf.toSheafify (PhysicsOfConsciousness.probabilityPresheaf_pre X)).app
      (Opposite.op U)))
  (PhysicsOfConsciousness.measureOnOpen μ U)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.globalSectionOfMeasure
PhysicsOfConsciousness.measureOnOpen
PhysicsOfConsciousness.measureOnOpen_restrict
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.probabilityPresheaf_pre

```

A.23 measure-patch-equality

PhysicsOfConsciousness.measureOnOpen_eq_of_restrict_eq

[Source, line 413](#); 49 nested tactic records.

```

∀ {X : TopCat} [inst : MeasurableSpace ↑X] [BorelSpace ↑X] {μ ν : MeasureTheory.FiniteMeasure ↑X}
  (U : TopologicalSpace.Opens ↑X),
  (↑μ).restrict ↑U = (↑ν).restrict ↑U →
  PhysicsOfConsciousness.measureOnOpen μ U = PhysicsOfConsciousness.measureOnOpen ν U

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.measureOnOpen

```

A.24 measure-representation

PhysicsOfConsciousness.ThermodynamicCover.existsUnique_measure_representation

[Source, line 434](#); 66 nested tactic records.

```

∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  (T : PhysicsOfConsciousness.ThermodynamicCover X)
  (μ : PhysicsOfConsciousness.LocalSectionSynchronization.I X → MeasureTheory.FiniteMeasure ↑X),
  (∀ (i j : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
    (↑(μ i)).restrict
      (↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover i) n
        ↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover j)) =
    (↑(μ j)).restrict
      (↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover i) n
        ↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover j))) →
  (∀ (i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
    PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i =
    (CategoryTheory.ConcreteCategory.hom
      ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ↪).op))
      (PhysicsOfConsciousness.globalSectionOfMeasure (μ i))) →
  ∃! ν,
  (∀ (i : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
    (↑ν).restrict ↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover i) =
    (↑(μ i)).restrict ↑(PhysicsOfConsciousness.LocalSectionSynchronization.cover i)) ∧
    PhysicsOfConsciousness.globalSectionOfMeasure ν = T.invariantMeasure

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.is_cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.SpatialMeasure.finite_glue_unique
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.I_fintype
PhysicsOfConsciousness.ThermodynamicCover.invariantMeasure
PhysicsOfConsciousness.ThermodynamicCover.invariantMeasure_unique
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.globalSectionOfMeasure
PhysicsOfConsciousness.globalSectionOfMeasure_restrict
PhysicsOfConsciousness.measureOnOpen
PhysicsOfConsciousness.measureOnOpen_eq_of_restrict_eq
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.probabilityPresheaf_pre

```

A.25 hasderivat-vmmoment

`PhysicsOfConsciousness.hasDerivAt_vmMoment`

[Source, line 34](#); 227 nested tactic records.

```

V (n : ℕ) (a : ℝ),
  HasDerivAt (PhysicsOfConsciousness.vmMoment n) (PhysicsOfConsciousness.vmMoment (n + 1) a) a

```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.vmMoment
```

A.26 vmmoment-zero-zero

`PhysicsOfConsciousness.vmMoment_zero_zero`

[Source, line 77](#); 11 nested tactic records.

```
PhysicsOfConsciousness.vmMoment 0 0 = 2 * π
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.vmMoment
```

A.27 vmmoment-one-zero

`PhysicsOfConsciousness.vmMoment_one_zero`

[Source, line 79](#); 5 nested tactic records.

```
PhysicsOfConsciousness.vmMoment 1 0 = 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.vmMoment
```

A.28 vmmoment-two-zero

`PhysicsOfConsciousness.vmMoment_two_zero`

[Source, line 81](#); 5 nested tactic records.

```
PhysicsOfConsciousness.vmMoment 2 0 = π
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.vmMoment
```

A.29 vmmoment-three-zero

PhysicsOfConsciousness.vmMoment_three_zero
[Source, line 83](#); 5 nested tactic records.

PhysicsOfConsciousness.vmMoment 3 0 = 0

Direct project constants in the proof term:

PhysicsOfConsciousness.vmMoment

A.30 vmmoment-four-zero

PhysicsOfConsciousness.vmMoment_four_zero
[Source, line 86](#); 10 nested tactic records.

PhysicsOfConsciousness.vmMoment 4 0 = 3 * π / 4

Direct project constants in the proof term:

PhysicsOfConsciousness.vmMoment

A.31 differentiable-vmmoment

PhysicsOfConsciousness.differentiable_vmMoment
[Source, line 91](#); 0 nested tactic records.

$\forall (n : \mathbb{N}), \text{Differentiable } \mathbb{R} (\text{PhysicsOfConsciousness.vmMoment } n)$

Direct project constants in the proof term:

PhysicsOfConsciousness.hasDerivAt_vmMoment
PhysicsOfConsciousness.vmMoment

A.32 deriv-vmmoment

PhysicsOfConsciousness.deriv_vmMoment
[Source, line 95](#); 11 nested tactic records.

$\forall (n : \mathbb{N}), \text{deriv } (\text{PhysicsOfConsciousness.vmMoment } n) = \text{PhysicsOfConsciousness.vmMoment } (n + 1)$

Direct project constants in the proof term:

PhysicsOfConsciousness.hasDerivAt_vmMoment
PhysicsOfConsciousness.vmMoment

A.33 contdiff-two-vmmoment

PhysicsOfConsciousness.contDiff_two_vmMoment
[Source, line 100](#); 70 nested tactic records.

$\forall (n : \mathbb{N}), \text{ContDiff } \mathbb{R}^2 (\text{PhysicsOfConsciousness.vmMoment } n)$

Direct project constants in the proof term:

PhysicsOfConsciousness.deriv_vmMoment
PhysicsOfConsciousness.differentiable_vmMoment
PhysicsOfConsciousness.vmMoment

A.34 vonmisessratio-eq-moments

PhysicsOfConsciousness.vonMisesSRatio_eq_moments

[Source, line 109](#); 231 nested tactic records.

```
∀ (a : ℝ),  
  PhysicsOfConsciousness.vonMisesSRatio a =  
    1 - PhysicsOfConsciousness.vmMoment 2 a / PhysicsOfConsciousness.vmMoment 0 a
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.vmMoment  
PhysicsOfConsciousness.vmMoment.eq_1  
PhysicsOfConsciousness.vonMisesS  
PhysicsOfConsciousness.vonMisesS.eq_1  
PhysicsOfConsciousness.vonMisesSRatio  
PhysicsOfConsciousness.vonMisesSRatio.eq_1  
PhysicsOfConsciousness.vonMisesWeight  
PhysicsOfConsciousness.vonMisesZ  
PhysicsOfConsciousness.vonMisesZ_pos
```

A.35 contdiff-two-vonmisessratio

PhysicsOfConsciousness.contDiff_two_vonMisesSRatio

[Source, line 129](#); 89 nested tactic records.

```
ContDiff ℝ 2 PhysicsOfConsciousness.vonMisesSRatio
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.contDiff_two_vmMoment  
PhysicsOfConsciousness.vmMoment  
PhysicsOfConsciousness.vonMisesSRatio  
PhysicsOfConsciousness.vonMisesSRatio_eq_moments  
PhysicsOfConsciousness.vonMisesZ  
PhysicsOfConsciousness.vonMisesZ_pos
```

A.36 hasderivat-vonmisessratio

PhysicsOfConsciousness.hasDerivAt_vonMisesSRatio

[Source, line 145](#); 93 nested tactic records.

```
∀ (a : ℝ), HasDerivAt PhysicsOfConsciousness.vonMisesSRatio (PhysicsOfConsciousness.sRatioDeriv a) a
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.hasDerivAt_vmMoment  
PhysicsOfConsciousness.sRatioDeriv  
PhysicsOfConsciousness.vmMoment  
PhysicsOfConsciousness.vonMisesSRatio  
PhysicsOfConsciousness.vonMisesSRatio_eq_moments  
PhysicsOfConsciousness.vonMisesZ  
PhysicsOfConsciousness.vonMisesZ_pos
```

A.37 sratioderiv-zero

PhysicsOfConsciousness.sRatioDeriv_zero

[Source, line 158](#); 5 nested tactic records.

```
PhysicsOfConsciousness.sRatioDeriv 0 = 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.sRatioDeriv  
PhysicsOfConsciousness.vmMoment
```

A.38 hasderivat-sratioderiv-zero

PhysicsOfConsciousness.hasDerivAt_sRatioDeriv_zero
[Source, line 162](#); 142 nested tactic records.

```
HasDerivAt PhysicsOfConsciousness.sRatioDeriv (-1 / 8) 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.hasDerivAt_vmMoment  
PhysicsOfConsciousness.sRatioDeriv  
PhysicsOfConsciousness.vmMoment
```

A.39 deriv-vonmisesratio

PhysicsOfConsciousness.deriv_vonMisesSRatio
[Source, line 183](#); 11 nested tactic records.

```
deriv PhysicsOfConsciousness.vonMisesSRatio = PhysicsOfConsciousness.sRatioDeriv
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.hasDerivAt_vonMisesSRatio  
PhysicsOfConsciousness.sRatioDeriv  
PhysicsOfConsciousness.vonMisesSRatio
```

A.40 deriv-sratioderiv-zero

PhysicsOfConsciousness.deriv_sRatioDeriv_zero
[Source, line 188](#); 0 nested tactic records.

```
deriv PhysicsOfConsciousness.sRatioDeriv 0 = -1 / 8
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.hasDerivAt_sRatioDeriv_zero  
PhysicsOfConsciousness.sRatioDeriv
```

A.41 second-order

PhysicsOfConsciousness.vonMisesSRatio_second_order
[Source, line 196](#); 200 nested tactic records.

```
(fun a => PhysicsOfConsciousness.vonMisesSRatio a - (1 / 2 - a ^ 2 / 16)) = 0 [N 0] fun a => a ^ 2
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.contDiff_two_vonMisesSRatio  
PhysicsOfConsciousness.deriv_sRatioDeriv_zero  
PhysicsOfConsciousness.deriv_vonMisesSRatio  
PhysicsOfConsciousness.sRatioDeriv  
PhysicsOfConsciousness.sRatioDeriv_zero  
PhysicsOfConsciousness.vonMisesSRatio  
PhysicsOfConsciousness.vonMisesSRatio_zero
```


A.45 branch-exponent

PhysicsOfConsciousness.coherentBranch_critical_exponent

[Source, line 317](#); 16 nested tactic records.

```
∀ {D : ℝ},
  0 < D →
  Filter.Tendsto (fun K => PhysicsOfConsciousness.coherentBranch D K ^ 2 / (K - 2 * D))
    (λ[>] (2 * D)) (λ (1 / D))
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.coherentBranch
PhysicsOfConsciousness.coherentBranch_spec
PhysicsOfConsciousness.coherentBranch_tendsto_zero
PhysicsOfConsciousness.coherent_solution_critical_exponent
PhysicsOfConsciousness.selfConsistency
```

A.46 unifiedself-self

PhysicsOfConsciousness.Chain.unifiedSelf_self

[Source, line 231](#); 6 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  {rb : PhysicsOfConsciousness.ReflexiveBoundary X},
  PhysicsOfConsciousness.Chain.UnifiedSelf rb → PhysicsOfConsciousness.Chain.Self rb
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.IsUnifiedBy
PhysicsOfConsciousness.Chain.Self
PhysicsOfConsciousness.Chain.UnifiedSelf
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.TriangulatedManifold
```

A.47 capacity

PhysicsOfConsciousness.Chain.capacity

[Source, line 243](#); 0 nested tactic records.

```
∀ (sys : Type u_1) [inst : Fintype sys] [Nonempty sys], PhysicsOfConsciousness.Chain.Capacity sys
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.is_prob_dist
PhysicsOfConsciousness.shannon_entropy_le_log_card
```

A.48 activebound-of-feedback

PhysicsOfConsciousness.Chain.activeBound_of_feedback

[Source, line 249](#); 0 nested tactic records.

```
∀ {Xs : Type u_1} {S : Type u_2} [inst : Fintype Xs] [inst_1 : Fintype S] [Nonempty S]
  (M : PhysicsOfConsciousness.FiniteFeedbackStep Xs S),
  M.Positive →
  ∀ (θ : ℝ),
  0 < θ →
  ∀ (q : Xs → S → S → ℝ),
  M.LocalDetailedBalance θ q →
  ∀ (budget : ℝ),
  M.meanHeat q ≤ budget → PhysicsOfConsciousness.Chain.ActiveBound M θ q budget
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.LocalDetailedBalance
PhysicsOfConsciousness.FiniteFeedbackStep.Positive
PhysicsOfConsciousness.FiniteFeedbackStep.final
PhysicsOfConsciousness.FiniteFeedbackStep.heat_bound
PhysicsOfConsciousness.FiniteFeedbackStep.initial
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.ProbDist.p
PhysicsOfConsciousness.shannon_entropy
```

A.49 activebound-of-continuing

PhysicsOfConsciousness.Chain.activeBound_of_continuing

[Source, line 260](#); 18 nested tactic records.

```
∀ {X : Type u_1} {S : Type u_2} [inst : Fintype X] [inst_1 : Fintype S] [Nonempty S]
  (C : PhysicsOfConsciousness.ContinuingProcess X S),
  C.protocol.Positive →
  0 < C.temperature →
  C.protocol.LocalDetailedBalance C.temperature C.heat →
  ∀ (N k : ℕ),
  k < N →
  C.Sustains N →
  (∀ j ∈ Finset.range N, 0 ≤ (C.protocol.step j).meanHeat (C.heat j)) →
  PhysicsOfConsciousness.Chain.ActiveBound (C.protocol.step k) C.temperature
  (C.heat k)
  (C.stored + ∑ z, (C.protocol.law 0).p z * C.energy z -
   ∑ z, (C.protocol.law N).p z * C.energy z)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.activeBound_of_feedback
PhysicsOfConsciousness.ContinuingProcess
PhysicsOfConsciousness.ContinuingProcess.Sustains
PhysicsOfConsciousness.ContinuingProcess.energy
PhysicsOfConsciousness.ContinuingProcess.heat
PhysicsOfConsciousness.ContinuingProcess.protocol
PhysicsOfConsciousness.ContinuingProcess.stored
PhysicsOfConsciousness.ContinuingProcess.temperature
PhysicsOfConsciousness.ContinuingProcess.totalHeat
PhysicsOfConsciousness.ContinuingProcess.totalHeat_le_stored
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.FiniteProtocol.LocalDetailedBalance
PhysicsOfConsciousness.FiniteProtocol.Positive
PhysicsOfConsciousness.FiniteProtocol.law
PhysicsOfConsciousness.FiniteProtocol.step
PhysicsOfConsciousness.FiniteProtocol.step_positive
PhysicsOfConsciousness.ProbDist.p
```

A.50 continuing-stage-activebound

PhysicsOfConsciousness.Chain.continuing_stage_activeBound

[Source, line 276](#); 157 nested tactic records.

```
∀ k < 3,
  PhysicsOfConsciousness.Chain.ActiveBound
  (PhysicsOfConsciousness.Examples.Continuing.agent.protocol.step k) 1
  (PhysicsOfConsciousness.Examples.Continuing.agent.protocol.stageHeat 1 k) (4 * Real.log 3)
```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.activeBound_of_continuing
PhysicsOfConsciousness.ContinuingAgent.protocol
PhysicsOfConsciousness.ContinuingProcess
PhysicsOfConsciousness.ContinuingProcess.energy
PhysicsOfConsciousness.ContinuingProcess.heat
PhysicsOfConsciousness.ContinuingProcess.mk
PhysicsOfConsciousness.ContinuingProcess.protocol
PhysicsOfConsciousness.ContinuingProcess.stored
PhysicsOfConsciousness.ContinuingProcess.temperature
PhysicsOfConsciousness.Examples.Continuing.act_cost
PhysicsOfConsciousness.Examples.Continuing.actuate._proof_1
PhysicsOfConsciousness.Examples.Continuing.actuate._proof_2
PhysicsOfConsciousness.Examples.Continuing.agent
PhysicsOfConsciousness.Examples.Continuing.first_cycle_sustained
PhysicsOfConsciousness.Examples.Continuing.law_zero_masses
PhysicsOfConsciousness.Examples.Continuing.learn_cost
PhysicsOfConsciousness.Examples.Continuing.log_ratio_pos
PhysicsOfConsciousness.Examples.Continuing.log_three_pos
PhysicsOfConsciousness.Examples.Continuing.process
PhysicsOfConsciousness.Examples.Continuing.process.eq_1
PhysicsOfConsciousness.Examples.Continuing.protocol_positive
PhysicsOfConsciousness.Examples.Continuing.reset_cost_pos
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.FiniteProtocol
PhysicsOfConsciousness.FiniteProtocol.law
PhysicsOfConsciousness.FiniteProtocol.local_balance
PhysicsOfConsciousness.FiniteProtocol.stageHeat
PhysicsOfConsciousness.FiniteProtocol.step
PhysicsOfConsciousness.ProbDist
PhysicsOfConsciousness.ProbDist.p

```

A.51 entropy-budget

PhysicsOfConsciousness.Chain.ActiveBound.entropy_budget

[Source, line 305](#); 7 nested tactic records.

```

∀ {Xs : Type u_1} {S : Type u_2} [inst : Fintype Xs] [inst_1 : Fintype S]
  {M : PhysicsOfConsciousness.FiniteFeedbackStep Xs S} {θ : ℝ} {q : Xs → S → S → ℝ} {budget : ℝ},
  PhysicsOfConsciousness.Chain.ActiveBound M θ q budget →
    PhysicsOfConsciousness.shannon_entropy M.initial.p -
      PhysicsOfConsciousness.shannon_entropy M.final.p ≤
      budget / θ

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.final
PhysicsOfConsciousness.FiniteFeedbackStep.initial
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.ProbDist.p
PhysicsOfConsciousness.shannon_entropy

```

A.52 leavesvacuum-of-separated

PhysicsOfConsciousness.Chain.leavesVacuum_of_separated

[Source, line 313](#); 0 nested tactic records.

```

∀ {Y : Type u_1} {V : Type u_2} [inst : TopologicalSpace Y] [PreconnectedSpace Y]
  [inst_2 : TopologicalSpace V] (vac : Set V) (phi : Y → V),
  Continuous phi →

```

```

  ∀ (y₁ y₂ : Y),
  (∀ (S : Set V), IsPreconnected S → S ⊆ vac → phi y₁ ∈ S → phi y₂ ∉ S) →
  PhysicsOfConsciousness.Chain.LeavesVacuum vac phi

```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.exists_notMem_of_no_common_preconnected
```

A.53 dissipates-of-not-surjective

PhysicsOfConsciousness.Chain.dissipates_of_not_surjective

[Source, line 321](#); 0 nested tactic records.

```

  ∀ {sys : Type u_1} [inst : Fintype sys] [inst_1 : DecidableEq sys] [Nonempty sys]
  [inst_3 : PhysicsOfConsciousness.StatisticalMechanics sys] (t : sys → sys),
  ¬Function.Surjective t → PhysicsOfConsciousness.Chain.Dissipates t

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.StatisticalMechanics
PhysicsOfConsciousness.finite_phase_space_dissipates

```

A.54 predictivebound-of-nonempty

PhysicsOfConsciousness.Chain.predictiveBound_of_nonempty

[Source, line 327](#); 0 nested tactic records.

```

  ∀ {Xs : Type u_1} {Sg : Type u_2} {Sg' : Type u_3} [inst : MeasurableSpace Xs]
  [inst_1 : MeasurableSpace Sg] [inst_2 : MeasurableSpace Sg'],
  Nonempty (PhysicsOfConsciousness.PredictiveDissipation Xs Sg Sg') →
  PhysicsOfConsciousness.Chain.PredictiveBound Xs Sg Sg'

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.PredictiveBound
PhysicsOfConsciousness.PredictiveDissipation
PhysicsOfConsciousness.PredictiveDissipation.dissipatedWork
PhysicsOfConsciousness.PredictiveDissipation.nonpredictive
PhysicsOfConsciousness.PredictiveDissipation.nonpredictive_le_dissipation
PhysicsOfConsciousness.PredictiveDissipation.thermalEnergy

```

A.55 coarsegrains-of-meshrefinement

PhysicsOfConsciousness.Chain.coarseGrains_of_meshRefinement

[Source, line 334](#); 0 nested tactic records.

```

  ∀ {M : Type u} [inst : PseudoMetricSpace M] [inst_1 : MeasurableSpace M]
  (μ : MeasureTheory.Measure M) (f : M → ℝ) {S : Set M},
  UniformContinuous f →
  μ S ≠ ∞ →
  MeasureTheory.IntegrableOn f S μ →
  ∀ (TM : ℕ → PhysicsOfConsciousness.TriangulatedManifold M)
  [inst_2 : (n : ℕ) → Fintype (PhysicsOfConsciousness.TriangulatedManifold.V M)]
  [inst_3 : (n : ℕ) → LinearOrder (PhysicsOfConsciousness.TriangulatedManifold.V M)],
  (∀ (n : ℕ), PhysicsOfConsciousness.IsRegularTriangulation (TM n) S) →
  ∀ (m : ℕ → ℝ),
  Filter.Tendsto m Filter.atTop (λ θ) →
  (∀ (n : ℕ) (u v : PhysicsOfConsciousness.TriangulatedManifold.V M),
   ∀ x ∈ PhysicsOfConsciousness.TriangulatedManifold.edge_region u v,
   ∀ y ∈ PhysicsOfConsciousness.TriangulatedManifold.edge_region u v,
   dist x y ≤ m n) →
  PhysicsOfConsciousness.Chain.CoarseGrains
  (fun n => PhysicsOfConsciousness.discreteEnergy (TM n) μ f)
  (λ (x : M) in S, f x ∂μ)

```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.IsRegularTriangulation
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.TriangulatedManifold.V
PhysicsOfConsciousness.TriangulatedManifold.edge_region
PhysicsOfConsciousness.mesh_refinement_convergence
```

A.56 coherent-of-supercritical

PhysicsOfConsciousness.Chain.coherent_of_supercritical
[Source, line 346](#); 0 nested tactic records.

```
∀ {K D : ℝ},
  0 < D → PhysicsOfConsciousness.critical_coupling D < K → PhysicsOfConsciousness.Chain.Coherent K D
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.supercritical_fixed_point_existsUnique
```

A.57 supercritical-of-coherent

PhysicsOfConsciousness.Chain.supercritical_of_coherent
[Source, line 362](#); 20 nested tactic records.

```
∀ {K D : ℝ},
  0 < D →
    0 ≤ K →
      PhysicsOfConsciousness.Chain.Coherent K D → PhysicsOfConsciousness.critical_coupling D < K
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.fixed_point_eq_zero_of_le_critical
PhysicsOfConsciousness.selfConsistency
```

A.58 unity-of-cover

PhysicsOfConsciousness.Chain.unity_of_cover
[Source, line 370](#); 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X],
  Nonempty (PhysicsOfConsciousness.ThermodynamicCover X) → PhysicsOfConsciousness.Chain.Unity X
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Unity
PhysicsOfConsciousness.Chain.Unity._proof_1
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.global_section_from_thermodynamics
PhysicsOfConsciousness.probabilityPresheaf
```

A.59 nonempty-globalsection-of-unity

PhysicsOfConsciousness.Chain.nonempty_globalSection_of_unity

[Source, line 376](#); 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X],
  PhysicsOfConsciousness.Chain.Unity X → Nonempty PhysicsOfConsciousness.GlobalSection
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Unity
PhysicsOfConsciousness.Chain.Unity._proof_1
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.probabilityPresheaf
```

A.60 self-of-contraction

PhysicsOfConsciousness.Chain.self_of_contraction

[Source, line 382](#); 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [Nonempty PhysicsOfConsciousness.GlobalSection]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection] {K D τ : ℝ},
  0 < τ →
  PhysicsOfConsciousness.critical_coupling D < K →
  ∀ (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
  LipschitzWith (PhysicsOfConsciousness.resonanceRate K D τ) rb.predict →
  PhysicsOfConsciousness.Chain.Self rb
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.resonanceRate
PhysicsOfConsciousness.self_of_supercritical
```

A.61 self-of-coherent-order-parameter

PhysicsOfConsciousness.Chain.self_of_coherent_order_parameter

[Source, line 400](#); 0 nested tactic records.

```
∀ {X : TopCat} [inst : MeasurableSpace ↑X] [inst_1 : BorelSpace ↑X]
  [Nonempty PhysicsOfConsciousness.GlobalSection]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection] {K D τ : ℝ},
  0 < D →
  0 ≤ K →
  0 < τ →
  PhysicsOfConsciousness.Chain.Coherent K D →
  ∀ (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
  LipschitzWith (PhysicsOfConsciousness.resonanceRate K D τ) rb.predict →
  PhysicsOfConsciousness.Chain.Self rb
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.Chain.self_of_contraction
PhysicsOfConsciousness.Chain.supercritical_of_coherent
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.resonanceRate
```

A.62 e45active-of-actuatedcoupling

PhysicsOfConsciousness.Chain.e45Active_of_actuatedCoupling

[Source, line 556](#); 0 nested tactic records.

```
∀ {M : Type u_1} {Xs : Type u_2} {S : Type u_3} [inst : PseudoMetricSpace M]
  [inst_1 : MeasurableSpace M] [inst_2 : Fintype Xs] [inst_3 : Fintype S]
  (A : PhysicsOfConsciousness.ActuatedCoupling M Xs S) (budget : ℝ),
  PhysicsOfConsciousness.Chain.E45Active A.step A.temperature A.heat budget A.energy
  A.continuumLimit
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ActuatedCoupling
PhysicsOfConsciousness.ActuatedCoupling.actuated_coarseGrains
PhysicsOfConsciousness.ActuatedCoupling.heat
PhysicsOfConsciousness.ActuatedCoupling.step
PhysicsOfConsciousness.ActuatedCoupling.temperature
PhysicsOfConsciousness.Chain.ActiveBound
```

A.63 actuated-limit-le-budget

PhysicsOfConsciousness.Chain.actuated_limit_le_budget

[Source, line 566](#); 0 nested tactic records.

```
∀ {M : Type u_1} {Xs : Type u_2} {S : Type u_3} [inst : PseudoMetricSpace M]
  [inst_1 : MeasurableSpace M] [inst_2 : Fintype Xs] [inst_3 : Fintype S]
  (A : PhysicsOfConsciousness.ActuatedCoupling M Xs S) {budget : ℝ},
  PhysicsOfConsciousness.Chain.ActiveBound A.step A.temperature A.heat budget →
  0 ≤ ∫ (x : M) in A.region, A.base x ∂A.volume →
  A.continuumLimit ≤
  A.gain * (budget / A.temperature) * ∫ (x : M) in A.region, A.base x ∂A.volume
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ActuatedCoupling
PhysicsOfConsciousness.ActuatedCoupling.actuated_limit_le_of_drive_le
PhysicsOfConsciousness.ActuatedCoupling.base
PhysicsOfConsciousness.ActuatedCoupling.heat
PhysicsOfConsciousness.ActuatedCoupling.region
PhysicsOfConsciousness.ActuatedCoupling.step
PhysicsOfConsciousness.ActuatedCoupling.temperature
PhysicsOfConsciousness.ActuatedCoupling.volume
PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.ActiveBound.entropy_budget
```

A.64 e45active-of-localactuator

PhysicsOfConsciousness.Chain.e45Active_of_localActuator

[Source, line 593](#); 0 nested tactic records.

```

V {M : Type u_1} {X : Type u_2} {S : Type u_3} {I : Type u_4} [inst : TopologicalSpace M]
  [inst_1 : CompactSpace M] [inst_2 : MeasurableSpace M] [inst_3 : BorelSpace M]
  [inst_4 : SecondCountableTopology M] [inst_5 : Fintype X] [inst_6 : Fintype S]
  [inst_7 : Fintype I] (A : PhysicsOfConsciousness.KernelArrangement M X S I)
  (P : PhysicsOfConsciousness.FiniteFeedbackStep X S),
A.law = P.final →
  V (θ : ℝ) (q : X → S → S → ℝ) (budget : ℝ),
    PhysicsOfConsciousness.Chain.E45Active P θ q budget A.energy A.continuumEnergy

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.final
PhysicsOfConsciousness.KernelArrangement
PhysicsOfConsciousness.KernelArrangement.coarseGrains
PhysicsOfConsciousness.KernelArrangement.law
PhysicsOfConsciousness.ProbDist

```

A.65 e45-of-localactuator

PhysicsOfConsciousness.Chain.e45_of_localActuator

[Source, line 616](#); 0 nested tactic records.

```

V {M : Type u_1} {X : Type u_2} {S : Type u_3} {Sg' : Type u_4} {I : Type u_5}
  [inst : TopologicalSpace M] [inst_1 : CompactSpace M] [inst_2 : MeasurableSpace M]
  [inst_3 : BorelSpace M] [inst_4 : SecondCountableTopology M] [inst_5 : Fintype X]
  [inst_6 : Fintype S] [inst_7 : Fintype I] [inst_8 : MeasurableSpace X]
  [MeasurableSingletonClass X] [inst_10 : MeasurableSpace S] [MeasurableSingletonClass S]
  [inst_12 : MeasurableSpace Sg'] (A : PhysicsOfConsciousness.KernelArrangement M X S I)
  (R : PhysicsOfConsciousness.PredictiveDissipation X S Sg'),
PhysicsOfConsciousness.PredictiveDissipation.joint Sg' = A.law.toMeasure →
  PhysicsOfConsciousness.Chain.E45 X S Sg' A.energy A.continuumEnergy

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.PredictiveBound
PhysicsOfConsciousness.KernelArrangement
PhysicsOfConsciousness.KernelArrangement.coarseGrains
PhysicsOfConsciousness.KernelArrangement.law
PhysicsOfConsciousness.PredictiveDissipation
PhysicsOfConsciousness.PredictiveDissipation.joint
PhysicsOfConsciousness.ProbDist.toMeasure

```

A.66 activebound-of-e34active

PhysicsOfConsciousness.Chain.activeBound_of_e34Active

[Source, line 629](#); 6 nested tactic records.

```

V {sys : Type u_1} {S : Type u_2} [inst : Fintype sys] [inst_1 : Fintype S] [Nonempty S]
  [inst_3 : PhysicsOfConsciousness.Thermodynamics sys] (upd : sys → sys)
  (M : PhysicsOfConsciousness.FiniteFeedbackStep sys S) (q : sys → S → S → ℝ),
PhysicsOfConsciousness.Chain.E34Active upd M q →
  PhysicsOfConsciousness.Chain.Dissipates upd →
    PhysicsOfConsciousness.Chain.ActiveBound M
      (PhysicsOfConsciousness.Thermodynamics.temperature sys) q
      (PhysicsOfConsciousness.heat_dissipation upd)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Chain.E34Active
PhysicsOfConsciousness.Chain.activeBound_of_feedback
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.LocalDetailedBalance
PhysicsOfConsciousness.FiniteFeedbackStep.Positive
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.Thermodynamics
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.Thermodynamics.temperature_pos
PhysicsOfConsciousness.heat_dissipation

```

A.67 e34active-of-ledger

PhysicsOfConsciousness.Chain.e34Active_of_ledger

[Source, line 649](#); 0 nested tactic records.

```

V {sys : Type u_1} {S : Type u_2} {n : ℕ} [inst : Fintype sys] [inst_1 : Fintype S] [Nonempty S]
  [inst_3 : PhysicsOfConsciousness.Thermodynamics sys]
  (L : PhysicsOfConsciousness.RegisterLedger sys S n) (a : Fin n),
  (∀ (i : Fin n), i ≠ a → L.Compressive i) →
    PhysicsOfConsciousness.Chain.E34Active L.update (L.step a) (L.heat a)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.FiniteFeedbackStep.LocalDetailedBalance
PhysicsOfConsciousness.FiniteFeedbackStep.Positive
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.RegisterLedger
PhysicsOfConsciousness.RegisterLedger.Compressive
PhysicsOfConsciousness.RegisterLedger.balance
PhysicsOfConsciousness.RegisterLedger.heat
PhysicsOfConsciousness.RegisterLedger.opHeat_le_dissipation
PhysicsOfConsciousness.RegisterLedger.positive
PhysicsOfConsciousness.RegisterLedger.step
PhysicsOfConsciousness.RegisterLedger.update
PhysicsOfConsciousness.Thermodynamics
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.heat_dissipation

```

A.68 dissipation-of-unreachable

PhysicsOfConsciousness.Chain.dissipation_of_unreachable

[Source, line 781](#); 0 nested tactic records.

```

V {sys : Type u_1} [inst : Fintype sys] [inst_1 : DecidableEq sys] [Nonempty sys]
  [inst_3 : PhysicsOfConsciousness.StatisticalMechanics sys] (t : sys → sys),
  ¬Function.Surjective t →
    PhysicsOfConsciousness.Chain.Capacity sys ∧ PhysicsOfConsciousness.Chain.Dissipates t

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.Capacity
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Chain.capacity
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.StatisticalMechanics
PhysicsOfConsciousness.StatisticalMechanics.toThermodynamics

```

A.69 downstream-chain

PhysicsOfConsciousness.Chain.chain_from_coarseGrains

Source, line 796; 91 nested tactic records.

```
∀ {X : TopCat} [inst : MeasureTheory.MeasureSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField ↑X) {E : ℕ → ℝ} {L K D τ : ℝ},
  PhysicsOfConsciousness.Chain.CoarseGrains E L →
  0 < τ →
  ∀ (T : PhysicsOfConsciousness.ThermodynamicCover X)
    (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
  PhysicsOfConsciousness.Chain.E56 kernel E L K D →
  PhysicsOfConsciousness.Chain.E67 L K D →
  PhysicsOfConsciousness.Chain.E78 K D T →
  PhysicsOfConsciousness.Chain.E89 K D τ T rb →
  PhysicsOfConsciousness.Chain.UnifiedSelf rb
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.E78
PhysicsOfConsciousness.Chain.E89
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.Chain.IsUnifiedBy
PhysicsOfConsciousness.Chain.Self
PhysicsOfConsciousness.Chain.UnifiedSelf
PhysicsOfConsciousness.Chain.Unity
PhysicsOfConsciousness.Chain.coherent_of_supercritical
PhysicsOfConsciousness.Chain.self_of_coherent_order_parameter
PhysicsOfConsciousness.Chain.unity_of_cover
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.resonanceRate
```

A.70 passive-chain

PhysicsOfConsciousness.Chain.chain

Source, line 858; 66 nested tactic records.

```
∀ {X : TopCat} [inst : MeasureTheory.MeasureSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField ↑X) {sys : Type u_1} [inst_5 : Fintype sys]
  [inst_6 : DecidableEq sys] [Nonempty sys]
  [inst_8 : PhysicsOfConsciousness.StatisticalMechanics sys] (upd : sys → sys) {Y : Type u_2}
  {V : Type u_3} (vac : Set V) (phi : Y → V) (Xs : Type u_4) (Sg : Type u_5) (Sg' : Type u_6)
  [inst_9 : MeasurableSpace Xs] [inst_10 : MeasurableSpace Sg] [inst_11 : MeasurableSpace Sg']
  (E : ℕ → ℝ) (L : ℝ) {K D τ : ℝ},
  0 < τ →
  ∀ (T : PhysicsOfConsciousness.ThermodynamicCover X)
```

```

(rb : PhysicsOfConsciousness.ReflexiveBoundary X),
PhysicsOfConsciousness.Chain.E12 sys vac phi →
  PhysicsOfConsciousness.Chain.E23 vac phi upd →
    PhysicsOfConsciousness.Chain.E34 upd Xs Sg Sg' →
      PhysicsOfConsciousness.Chain.E45 Xs Sg Sg' E L →
        PhysicsOfConsciousness.Chain.E56 kernel E L K D →
          PhysicsOfConsciousness.Chain.E67 L K D →
            PhysicsOfConsciousness.Chain.E78 K D T →
              PhysicsOfConsciousness.Chain.E89 K D τ T rb →
                PhysicsOfConsciousness.Chain.UnifiedSelf rb

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.Capacity
PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Chain.E12
PhysicsOfConsciousness.Chain.E23
PhysicsOfConsciousness.Chain.E34
PhysicsOfConsciousness.Chain.E45
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.E78
PhysicsOfConsciousness.Chain.E89
PhysicsOfConsciousness.Chain.LeavesVacuum
PhysicsOfConsciousness.Chain.PredictiveBound
PhysicsOfConsciousness.Chain.UnifiedSelf
PhysicsOfConsciousness.Chain.capacity
PhysicsOfConsciousness.Chain.chain_from_coarseGrains
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.Chain.predictiveBound_of_nonempty
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.PredictiveDissipation
PhysicsOfConsciousness.PredictiveDissipation.dissipatedWork
PhysicsOfConsciousness.PredictiveDissipation.nonpredictive
PhysicsOfConsciousness.PredictiveDissipation.thermalEnergy
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.StatisticalMechanics
PhysicsOfConsciousness.StatisticalMechanics.toThermodynamics
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.heat_dissipation

```

A.71 chain-active

PhysicsOfConsciousness.Chain.chain_active

[Source, line 912](#); 53 nested tactic records.

```

∀ {X : TopCat} [inst : MeasureTheory.MeasureSpace ↑X] [inst_1 : BorelSpace ↑X]
  [inst_2 : PhysicsOfConsciousness.TriangulatedManifold ↑X]
  [inst_3 : MetricSpace PhysicsOfConsciousness.GlobalSection]
  [CompleteSpace PhysicsOfConsciousness.GlobalSection]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField ↑X) {sys : Type u_1} [inst_5 : Fintype sys]
  [inst_6 : DecidableEq sys] [Nonempty sys]
  [inst_8 : PhysicsOfConsciousness.StatisticalMechanics sys] (upd : sys → sys) {Y : Type u_2}
  {V : Type u_3} (vac : Set V) (phi : Y → V) {S : Type u_4} [inst_9 : Fintype S] [Nonempty S]
  (M : PhysicsOfConsciousness.FiniteFeedbackStep sys S) (q : sys → S → S → ℝ) (E : ℕ → ℝ) (L : ℝ)
  {K D τ : ℝ},
  0 < τ →
  ∀ (T : PhysicsOfConsciousness.ThermodynamicCover X)
    (rb : PhysicsOfConsciousness.ReflexiveBoundary X),
    PhysicsOfConsciousness.Chain.E12 sys vac phi →

```

```

PhysicsOfConsciousness.Chain.E23 vac phi upd →
PhysicsOfConsciousness.Chain.E34Active upd M q →
PhysicsOfConsciousness.Chain.E45Active M
  (PhysicsOfConsciousness.Thermodynamics.temperature sys) q
  (PhysicsOfConsciousness.heat_dissipation upd) E L →
PhysicsOfConsciousness.Chain.E56 kernel E L K D →
PhysicsOfConsciousness.Chain.E67 L K D →
PhysicsOfConsciousness.Chain.E78 K D T →
PhysicsOfConsciousness.Chain.E89 K D τ T rb →
PhysicsOfConsciousness.Chain.UnifiedSelf rb

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.Capacity
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Chain.E12
PhysicsOfConsciousness.Chain.E23
PhysicsOfConsciousness.Chain.E34Active
PhysicsOfConsciousness.Chain.E45Active
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.E78
PhysicsOfConsciousness.Chain.E89
PhysicsOfConsciousness.Chain.LeavesVacuum
PhysicsOfConsciousness.Chain.activeBound_of_e34Active
PhysicsOfConsciousness.Chain.capacity
PhysicsOfConsciousness.Chain.chain_from_coarseGrains
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.ReflexiveBoundary
PhysicsOfConsciousness.StatisticalMechanics
PhysicsOfConsciousness.StatisticalMechanics.toThermodynamics
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.ThermodynamicCover
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.heat_dissipation

```

A.72 coherent-three-one

PhysicsOfConsciousness.Chain.coherent_three_one
[Source, line 950](#); 34 nested tactic records.

```
PhysicsOfConsciousness.Chain.Coherent 3 1
```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.coherent_of_supercritical
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1
PhysicsOfConsciousness.critical_coupling.eq_1

```

A.73 e67-three-one

PhysicsOfConsciousness.Chain.e67_three_one
[Source, line 956](#); 34 nested tactic records.

```
PhysicsOfConsciousness.Chain.E67 3 3 1
```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1
PhysicsOfConsciousness.critical_coupling.eq_1

```

A.74 e56-of-eq

PhysicsOfConsciousness.Chain.e56_of_eq

[Source, line 962](#); 5 nested tactic records.

```

∀ {M : Type u_1} [inst : MeasureTheory.MeasureSpace M] [inst_1 : TopologicalSpace M]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField M) (E : ℕ → ℝ),
  Nonempty (PhysicsOfConsciousness.IsEMFieldCoupling kernel 3 1) →
  PhysicsOfConsciousness.Chain.E56 kernel E 3 3 1

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.StochasticNeuralField

```

A.75 em-field-exhibits-phase-transition

PhysicsOfConsciousness.Chain.em_field_exhibits_phase_transition

[Source, line 976](#); 0 nested tactic records.

```

∀ {M : Type u_1} [inst : MeasureTheory.MeasureSpace M] [inst_1 : TopologicalSpace M]
  [inst_2 : MeasureTheory.IsProbabilityMeasure MeasureTheory.volume]
  (kernel : PhysicsOfConsciousness.StochasticNeuralField M) {E : ℕ → ℝ} {L K D : ℝ},
  PhysicsOfConsciousness.Chain.E56 kernel E L K D →
  PhysicsOfConsciousness.Chain.E67 L K D →
  PhysicsOfConsciousness.Chain.CoarseGrains E L →
  PhysicsOfConsciousness.exhibits_phase_transition kernel

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.exhibits_phase_transition_of_isEMFieldCoupling

```

A.76 installedenergy-required-by-chain

PhysicsOfConsciousness.Chain.installedEnergy_required_by_chain

[Source, line 995](#); 103 nested tactic records.

```

∀ {M : Type u_1} {X : Type u_2} {S : Type u_3} {I : Type u_4} [inst : MeasureTheory.MeasureSpace M]
  [inst_1 : TopologicalSpace M] [inst_2 : CompactSpace M] [inst_3 : BorelSpace M]
  [inst_4 : SecondCountableTopology M] [inst_5 : Fintype X] [inst_6 : Fintype S]
  [inst_7 : Fintype I] (A : PhysicsOfConsciousness.PricedArrangement M X S I)
  (sys : PhysicsOfConsciousness.StochasticNeuralField M),
  A.volume = MeasureTheory.volume →
  sys.K = A.kernel →
  ∀ {K D : ℝ},
  PhysicsOfConsciousness.Chain.E56 sys A.energy A.continuumEnergy K D →
  PhysicsOfConsciousness.Chain.E67 A.continuumEnergy K D →
  PhysicsOfConsciousness.critical_coupling D / A.couplingPerEnergy < A.installedEnergy

```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.Chain.em_field_exhibits_phase_transition
PhysicsOfConsciousness.ContinuousNeuralField.K
PhysicsOfConsciousness.FokkerPlanck.incoherentRate
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.IsEMFieldCoupling.domain_probability
PhysicsOfConsciousness.IsEMFieldCoupling.noise_is_field_noise
PhysicsOfConsciousness.KernelArrangement.coarseGrains
PhysicsOfConsciousness.KernelArrangement.continuumEnergy
PhysicsOfConsciousness.KernelArrangement.energy
PhysicsOfConsciousness.KernelArrangement.installedEnergy
PhysicsOfConsciousness.KernelArrangement.kernel
PhysicsOfConsciousness.KernelArrangement.volume
PhysicsOfConsciousness.PricedArrangement
PhysicsOfConsciousness.PricedArrangement.couplingPerEnergy
PhysicsOfConsciousness.PricedArrangement.couplingPerEnergy_pos
PhysicsOfConsciousness.PricedArrangement.no_coherence_of_installedEnergy
PhysicsOfConsciousness.PricedArrangement.toKernelArrangement
PhysicsOfConsciousness.StochasticNeuralField
PhysicsOfConsciousness.StochasticNeuralField.D
PhysicsOfConsciousness.StochasticNeuralField.toContinuousNeuralField
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.exhibits_phase_transition
PhysicsOfConsciousness.mean_field_coupling
PhysicsOfConsciousness.selfConsistency
```

A.77 below-installation-rejects-supercritical-edge

PhysicsOfConsciousness.Chain.below_installation_rejects_supercritical_edge

[Source, line 1022](#); 47 nested tactic records.

```
∀ {K : ℝ},
  PhysicsOfConsciousness.Chain.E56 PhysicsOfConsciousness.Examples.InstalledCoupling.belowField
    PhysicsOfConsciousness.Examples.InstalledCoupling.below.energy
    PhysicsOfConsciousness.Examples.InstalledCoupling.below.continuumEnergy K 1 →
  ¬PhysicsOfConsciousness.Chain.E67
    PhysicsOfConsciousness.Examples.InstalledCoupling.below.continuumEnergy K 1
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E56
PhysicsOfConsciousness.Chain.E67
PhysicsOfConsciousness.Chain.installedEnergy_required_by_chain
PhysicsOfConsciousness.ContinuousNeuralField.K
PhysicsOfConsciousness.Examples.InstalledCoupling.below
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_1
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_2
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_3
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_4
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_5
PhysicsOfConsciousness.Examples.InstalledCoupling.below._proof_6
PhysicsOfConsciousness.Examples.InstalledCoupling.belowField
PhysicsOfConsciousness.Examples.InstalledCoupling.belowLaw
PhysicsOfConsciousness.Examples.InstalledCoupling.below_installedEnergy
PhysicsOfConsciousness.Examples.InstalledCoupling.priced
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_1
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_2
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_3
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_4
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_5
```

```

PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_6
PhysicsOfConsciousness.Examples.InstalledCoupling.priced._proof_7
PhysicsOfConsciousness.Examples.MicroscopicCoupling.installed
PhysicsOfConsciousness.KernelArrangement
PhysicsOfConsciousness.KernelArrangement.actuator
PhysicsOfConsciousness.KernelArrangement.continuumEnergy
PhysicsOfConsciousness.KernelArrangement.energy
PhysicsOfConsciousness.KernelArrangement.grid
PhysicsOfConsciousness.KernelArrangement.installedEnergy
PhysicsOfConsciousness.KernelArrangement.mk
PhysicsOfConsciousness.KernelArrangement.modeMass
PhysicsOfConsciousness.KernelArrangement.volume
PhysicsOfConsciousness.KernelMesh
PhysicsOfConsciousness.KernelMesh.snap
PhysicsOfConsciousness.LocalActuator
PhysicsOfConsciousness.LocalActuator.occupancy
PhysicsOfConsciousness.LocalActuator.price
PhysicsOfConsciousness.LocalActuator.profile
PhysicsOfConsciousness.PricedArrangement
PhysicsOfConsciousness.PricedArrangement.couplingPerEnergy
PhysicsOfConsciousness.PricedArrangement.mk
PhysicsOfConsciousness.PricedArrangement.mk.congr_simp
PhysicsOfConsciousness.PricedArrangement.toKernelArrangement
PhysicsOfConsciousness.StochasticNeuralField.toContinuousNeuralField
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1

```

A.78 e34-booleraser

PhysicsOfConsciousness.Chain.e34_boolEraser

[Source, line 1044](#); 0 nested tactic records.

PhysicsOfConsciousness.Chain.E34 (fun x => true) Bool Bool Bool

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Examples.boolThermo
PhysicsOfConsciousness.Examples.landauerSystem
PhysicsOfConsciousness.Examples.landauerSystem_nonpredictive_pos
PhysicsOfConsciousness.PredictiveDissipation
PhysicsOfConsciousness.PredictiveDissipation.dissipatedWork
PhysicsOfConsciousness.PredictiveDissipation.nonpredictive
PhysicsOfConsciousness.PredictiveDissipation.thermalEnergy
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.heat_dissipation

```

A.79 coarsegrains-const

PhysicsOfConsciousness.Chain.coarseGrains_const

[Source, line 1049](#); 0 nested tactic records.

$\forall (L : \mathbb{R}), \text{PhysicsOfConsciousness.Chain.CoarseGrains } (\text{fun } x \Rightarrow L) L$

A.80 cortexcoherent

PhysicsOfConsciousness.Chain.cortexCoherent

[Source, line 1067](#); 0 nested tactic records.

PhysicsOfConsciousness.Chain.Coherent 3 1

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.coherent_three_one

A.81 cortexhasself-of-coherent

PhysicsOfConsciousness.Chain.cortexHasSelf_of_coherent
[Source, line 1073](#); 5 nested tactic records.

```
PhysicsOfConsciousness.Chain.Self PhysicsOfConsciousness.Examples.cortexReflexive
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.cortexCoherent  
PhysicsOfConsciousness.Chain.self_of_coherent_order_parameter  
PhysicsOfConsciousness.Examples.Cortex  
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate  
PhysicsOfConsciousness.Examples.cortexReflexive  
PhysicsOfConsciousness.Examples.cortexTau  
PhysicsOfConsciousness.Examples.cortexTau_pos  
PhysicsOfConsciousness.Examples.gsComplete  
PhysicsOfConsciousness.Examples.gsMetric  
PhysicsOfConsciousness.Examples.instBorelSpaceSite  
PhysicsOfConsciousness.Examples.instMeasurableSpaceSite  
PhysicsOfConsciousness.Examples.instNonemptyGlobalSection
```

A.82 cortexhasself-of-coherent-eq

PhysicsOfConsciousness.Chain.cortexHasSelf_of_coherent_eq
[Source, line 1082](#); 0 nested tactic records.

```
∀ (s : PhysicsOfConsciousness.GlobalSection),  
  PhysicsOfConsciousness.Examples.cortexReflexive.predict s = s →  
    s = PhysicsOfConsciousness.Examples.cortexState
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Examples.Cortex  
PhysicsOfConsciousness.Examples.cortexPredict_fixed_unique  
PhysicsOfConsciousness.Examples.cortexReflexive  
PhysicsOfConsciousness.Examples.instBorelSpaceSite  
PhysicsOfConsciousness.Examples.instMeasurableSpaceSite  
PhysicsOfConsciousness.GlobalSection  
PhysicsOfConsciousness.ReflexiveBoundary.predict
```

A.83 witness-not-surjective

PhysicsOfConsciousness.Chain.witness_not_surjective
[Source, line 1108](#); 8 nested tactic records.

```
¬Function.Surjective fun x => true
```

A.84 cortexvolume-apply

PhysicsOfConsciousness.Chain.cortexVolume_apply
[Source, line 1128](#); 0 nested tactic records.

```
∀ (s : Set PhysicsOfConsciousness.Examples.Site),  
  MeasureTheory.volume s = 3-1 * MeasureTheory.Measure.count s
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.cortexMeasureSpace  
PhysicsOfConsciousness.Examples.Cortex  
PhysicsOfConsciousness.Examples.Site
```

A.85 instisprobabilitymeasurecarriercortexvolume

PhysicsOfConsciousness.Chain.instIsProbabilityMeasureCarrierCortexVolume
[Source, line 1132](#); 82 nested tactic records.

MeasureTheory.IsProbabilityMeasure MeasureTheory.volume

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexVolume_apply
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.Site
PhysicsOfConsciousness.Examples.instFintypeSite
PhysicsOfConsciousness.Examples.instMeasurableSpaceSite

A.86 cortexvolume-singleton-le

PhysicsOfConsciousness.Chain.cortexVolume_singleton_le
[Source, line 1142](#); 97 nested tactic records.

$\forall (y : \text{PhysicsOfConsciousness.Examples.Site}), (\text{MeasureTheory.volume } \{y\}).\text{toReal} \leq 1 / 2$

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexVolume_apply
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.Site
PhysicsOfConsciousness.Examples.instMeasurableSingletonClassSite
PhysicsOfConsciousness.Examples.instMeasurableSpaceSite

A.87 cortexneuralfield-isemfieldcoupling

PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
[Source, line 1162](#); 5 nested tactic records.

Nonempty
(PhysicsOfConsciousness.IsEMFieldCoupling PhysicsOfConsciousness.Chain.cortexNeuralField 3 1)

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexMeasureSpace._proof_2
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexVolume_singleton_le
PhysicsOfConsciousness.Chain.instIsProbabilityMeasureCarrierCortexVolume
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.instTopologicalSpaceSite
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.isEMFieldCoupling_const

A.88 cortexneuralfield-exhibits-phase-transition

PhysicsOfConsciousness.Chain.cortexNeuralField_exhibits_phase_transition
[Source, line 1170](#); 34 nested tactic records.

PhysicsOfConsciousness.exhibits_phase_transition PhysicsOfConsciousness.Chain.cortexNeuralField

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
PhysicsOfConsciousness.Chain.instIsProbabilityMeasureCarrierCortexVolume
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.instTopologicalSpaceSite
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1
PhysicsOfConsciousness.critical_coupling.eq_1
PhysicsOfConsciousness.exhibits_phase_transition_of_isEMFieldCoupling

```

A.89 not-isemfieldcoupling-of-zero-kernel

PhysicsOfConsciousness.Chain.not_isEMFieldCoupling_of_zero_kernel

[Source, line 1182](#); 47 nested tactic records.

```

¬PhysicsOfConsciousness.IsEMFieldCoupling (PhysicsOfConsciousness.constField 0 1 ...) 3 1

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.instIsProbabilityMeasureCarrierCortexVolume
PhysicsOfConsciousness.ContinuousNeuralField.K
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.instTopologicalSpaceSite
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.IsEMFieldCoupling.coupling_is_mean_field
PhysicsOfConsciousness.StochasticNeuralField.toContinuousNeuralField
PhysicsOfConsciousness.constField
PhysicsOfConsciousness.mean_field_coupling
PhysicsOfConsciousness.mean_field_coupling_const

```

A.90 t5-e12-doublewell

PhysicsOfConsciousness.Chain.t5_e12_doubleWell

[Source, line 1197](#); 0 nested tactic records.

```

PhysicsOfConsciousness.Chain.E12 Bool
  (PhysicsOfConsciousness.DynamicalVacuum PhysicsOfConsciousness.Examples.wellV)
  PhysicsOfConsciousness.Examples.kink

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.Capacity
PhysicsOfConsciousness.Examples.kink_leaves_vacuum

```

A.91 t5-e23-absorbingregister

PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister

[Source, line 1204](#); 0 nested tactic records.

```

PhysicsOfConsciousness.Chain.E23
  (PhysicsOfConsciousness.DynamicalVacuum PhysicsOfConsciousness.Examples.wellV)
  PhysicsOfConsciousness.Examples.kink fun x => true

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.LeavesVacuum
PhysicsOfConsciousness.Chain.witness_not_surjective
PhysicsOfConsciousness.DynamicalVacuum
PhysicsOfConsciousness.Examples.kink
PhysicsOfConsciousness.Examples.wellV

```

A.92 t5-refiningenergy-tendsto

PhysicsOfConsciousness.Chain.t5_refiningEnergy_tendsto
[Source, line 1220](#); 79 nested tactic records.

```
Filter.Tendsto PhysicsOfConsciousness.Chain.t5_refiningEnergy Filter.atTop (λ 3)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.t5_refiningEnergy  
PhysicsOfConsciousness.Examples.gridTriangulation  
PhysicsOfConsciousness.Examples.grid_mesh_refinement  
PhysicsOfConsciousness.Examples.instFintypeGridV  
PhysicsOfConsciousness.Examples.tent_uniformContinuous  
PhysicsOfConsciousness.discreteEnergy
```

A.93 t5-refiningenergy-moves

PhysicsOfConsciousness.Chain.t5_refiningEnergy_moves
[Source, line 1233](#); 37 nested tactic records.

```
PhysicsOfConsciousness.Chain.t5_refiningEnergy 0 ≠ PhysicsOfConsciousness.Chain.t5_refiningEnergy 1
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.cortexMeasureSpace._proof_2  
PhysicsOfConsciousness.Chain.t5_refiningEnergy  
PhysicsOfConsciousness.Chain.t5_refiningEnergy._proof_1  
PhysicsOfConsciousness.Examples.gridTriangulation  
PhysicsOfConsciousness.Examples.instFintypeGridV  
PhysicsOfConsciousness.Examples.tent_energy_one  
PhysicsOfConsciousness.Examples.tent_energy_two  
PhysicsOfConsciousness.discreteEnergy
```

A.94 t5-e45-refiningmesh

PhysicsOfConsciousness.Chain.t5_e45_refiningMesh
[Source, line 1242](#); 0 nested tactic records.

```
PhysicsOfConsciousness.Chain.E45 Bool Bool Bool PhysicsOfConsciousness.Chain.t5_refiningEnergy 3
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.PredictiveBound  
PhysicsOfConsciousness.Chain.t5_refiningEnergy_tendsto
```

A.95 chain-hypotheses-jointly-satisfiable

PhysicsOfConsciousness.Chain.chain_hypotheses_jointly_satisfiable
[Source, line 1267](#); 39 nested tactic records.

```
PhysicsOfConsciousness.Chain.UnifiedSelf PhysicsOfConsciousness.Examples.cortexReflexive
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.CoarseGrains  
PhysicsOfConsciousness.Chain.Coherent  
PhysicsOfConsciousness.Chain.FieldRealizes  
PhysicsOfConsciousness.Chain.IsUnifiedBy  
PhysicsOfConsciousness.Chain.Unity._proof_1  
PhysicsOfConsciousness.Chain.chain
```

```

PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
PhysicsOfConsciousness.Chain.e34_boolEraser
PhysicsOfConsciousness.Chain.t5_e12_doubleWell
PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister
PhysicsOfConsciousness.Chain.t5_e45_refiningMesh
PhysicsOfConsciousness.Chain.t5_refiningEnergy
PhysicsOfConsciousness.DynamicalVacuum
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.cortexCover
PhysicsOfConsciousness.Examples.cortexCover_reachedByRelaxation_three
PhysicsOfConsciousness.Examples.cortexPredict_fixed
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate
PhysicsOfConsciousness.Examples.cortexReflexive
PhysicsOfConsciousness.Examples.cortexState
PhysicsOfConsciousness.Examples.cortexTau
PhysicsOfConsciousness.Examples.cortexTau_pos
PhysicsOfConsciousness.Examples.gsComplete
PhysicsOfConsciousness.Examples.gsMetric
PhysicsOfConsciousness.Examples.instBorelSpaceSite
PhysicsOfConsciousness.Examples.instTriangulatedManifoldCarrierCortex
PhysicsOfConsciousness.Examples.kink
PhysicsOfConsciousness.Examples.wellV
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.critical_coupling
PhysicsOfConsciousness.critical_coupling._proof_1
PhysicsOfConsciousness.critical_coupling.eq_1
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.resonanceRate

```

A.96 continuousneuralfield-free

PhysicsOfConsciousness.Chain.continuousNeuralField_free

[Source, line 1312](#); 0 nested tactic records.

```

∀ {M : Type u_1} [inst : MeasureTheory.MeasureSpace M] [inst_1 : TopologicalSpace M] (omega : M → ℝ)
  (K : M → M → ℝ) (tau : ℝ), Nonempty (PhysicsOfConsciousness.ContinuousNeuralField M)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ContinuousNeuralField
PhysicsOfConsciousness.ContinuousNeuralField.mk

```

A.97 vertexkernel-sitedon

PhysicsOfConsciousness.Chain.vertexKernel_sitedOn

[Source, line 1324](#); 28 nested tactic records.

```

∀ {M : Type u_1} [inst : TopologicalSpace M] [inst_1 : DecidableEq M]
  (TM : PhysicsOfConsciousness.TriangulatedManifold M)
  [inst_2 : Fintype (PhysicsOfConsciousness.TriangulatedManifold.V M)]
  (w :
    PhysicsOfConsciousness.TriangulatedManifold.V M →
    PhysicsOfConsciousness.TriangulatedManifold.V M → ℝ),
  PhysicsOfConsciousness.SitedOn

```

```
(Finset.image PhysicsOfConsciousness.TriangulatedManifold.embedding Finset.univ)
(PhysicsOfConsciousness.Chain.vertexKernel TM w)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.TriangulatedManifold.V
PhysicsOfConsciousness.TriangulatedManifold.embedding
```

A.98 vertexkernel-apply-embedding

PhysicsOfConsciousness.Chain.vertexKernel_apply_embedding

[Source, line 1337](#); 132 nested tactic records.

```
∀ {M : Type u_1} [inst : TopologicalSpace M] [inst_1 : DecidableEq M]
(TM : PhysicsOfConsciousness.TriangulatedManifold M)
[inst_2 : Fintype (PhysicsOfConsciousness.TriangulatedManifold.V M)]
[DecidableEq (PhysicsOfConsciousness.TriangulatedManifold.V M)],
Function.Injective PhysicsOfConsciousness.TriangulatedManifold.embedding →
  ∀
    (w :
      PhysicsOfConsciousness.TriangulatedManifold.V M →
        PhysicsOfConsciousness.TriangulatedManifold.V M → ℝ)
    (u v : PhysicsOfConsciousness.TriangulatedManifold.V M),
    PhysicsOfConsciousness.Chain.vertexKernel TM w
      (PhysicsOfConsciousness.TriangulatedManifold.embedding u)
      (PhysicsOfConsciousness.TriangulatedManifold.embedding v) =
      w u v
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.vertexKernel
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.TriangulatedManifold.V
PhysicsOfConsciousness.TriangulatedManifold.embedding
```

A.99 vertexkernel-fieldcorrelation-eq-zero

PhysicsOfConsciousness.Chain.vertexKernel_fieldCorrelation_eq_zero

[Source, line 1369](#); 0 nested tactic records.

```
∀ {M : Type u_1} [inst : TopologicalSpace M] [inst_1 : MeasurableSpace M] [inst_2 : DecidableEq M]
(TM : PhysicsOfConsciousness.TriangulatedManifold M)
[inst_3 : Fintype (PhysicsOfConsciousness.TriangulatedManifold.V M)]
(w :
  PhysicsOfConsciousness.TriangulatedManifold.V M →
    PhysicsOfConsciousness.TriangulatedManifold.V M → ℝ)
(μ : MeasureTheory.Measure M) [MeasureTheory.NullSingletonClass μ] (theta : M → ℝ),
PhysicsOfConsciousness.fieldCorrelation μ theta (PhysicsOfConsciousness.Chain.vertexKernel TM w) =
  0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.vertexKernel
PhysicsOfConsciousness.Chain.vertexKernel_sitedOn
PhysicsOfConsciousness.TriangulatedManifold
PhysicsOfConsciousness.TriangulatedManifold.V
PhysicsOfConsciousness.TriangulatedManifold.embedding
PhysicsOfConsciousness.fieldCorrelation_sited_eq_zero
```

A.100 active-entropy-budget

PhysicsOfConsciousness.Chain.active_entropy_budget

[Source, line 1381](#); 0 nested tactic records.

```
∀ {X : Type u_1} {S : Type u_2} [inst : Fintype X] [inst_1 : Fintype S] [Nonempty S]
(M : PhysicsOfConsciousness.FiniteFeedbackStep X S),
M.Positive →
  ∀ (θ : ℝ),
  θ < 0 →
    ∀ (q : X → S → ℝ),
    M.LocalDetailedBalance θ q →
      ∀ (budget : ℝ),
      M.meanHeat q ≤ budget →
        PhysicsOfConsciousness.shannon_entropy M.initial.p -
          PhysicsOfConsciousness.shannon_entropy M.final.p ≤
          budget / θ
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.ActiveBound.entropy_budget
PhysicsOfConsciousness.Chain.activeBound_of_feedback
PhysicsOfConsciousness.FiniteFeedbackStep
PhysicsOfConsciousness.FiniteFeedbackStep.LocalDetailedBalance
PhysicsOfConsciousness.FiniteFeedbackStep.Positive
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
```

A.101 thermalagency-e34active

PhysicsOfConsciousness.Chain.thermalAgency_e34Active

[Source, line 1394](#); 102 nested tactic records.

```
PhysicsOfConsciousness.Chain.E34Active (fun x => true)
  PhysicsOfConsciousness.Examples.ThermalAgency.actuation
  PhysicsOfConsciousness.Examples.ThermalAgency.heat
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.Dissipates
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_cost_positive
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_entropy_production
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_heat
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_local_balance
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_positive
PhysicsOfConsciousness.Examples.ThermalAgency.heat
PhysicsOfConsciousness.Examples.boolThermo
PhysicsOfConsciousness.FiniteFeedbackStep.LocalDetailedBalance
PhysicsOfConsciousness.FiniteFeedbackStep.Positive
PhysicsOfConsciousness.FiniteFeedbackStep.entropyProduction
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.heat_dissipation
```

A.102 thermalagency-activebound

PhysicsOfConsciousness.Chain.thermalAgency_activeBound

[Source, line 1410](#); 0 nested tactic records.

```
PhysicsOfConsciousness.Chain.ActiveBound PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat (Real.log 2)
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.activeBound_of_e34Active
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.Chain.thermalAgency_e34Active
PhysicsOfConsciousness.Chain.witness_not_surjective
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.boolThermo
```

A.103 thermalagency-zero-budget-rejected

PhysicsOfConsciousness.Chain.thermalAgency_zero_budget_rejected

[Source, line 1417](#); 7 nested tactic records.

```
¬PhysicsOfConsciousness.Chain.ActiveBound PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.actuation_cost_positive
PhysicsOfConsciousness.Examples.ThermalAgency.heat
PhysicsOfConsciousness.FiniteFeedbackStep.entropyProduction
PhysicsOfConsciousness.FiniteFeedbackStep.final
PhysicsOfConsciousness.FiniteFeedbackStep.initial
PhysicsOfConsciousness.FiniteFeedbackStep.meanHeat
PhysicsOfConsciousness.ProbDist.p
PhysicsOfConsciousness.shannon_entropy
```

A.104 thermalagency-wrong-limit-rejected

PhysicsOfConsciousness.Chain.thermalAgency_wrong_limit_rejected

[Source, line 1424](#); 31 nested tactic records.

```
¬PhysicsOfConsciousness.Chain.E45Active PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat (Real.log 2) (fun x => 0) 3
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E45Active
PhysicsOfConsciousness.Chain.thermalAgency_activeBound
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
```

A.105 thermalagency-e45active

PhysicsOfConsciousness.Chain.thermalAgency_e45Active

[Source, line 1435](#); 0 nested tactic records.

```
PhysicsOfConsciousness.Chain.E45Active PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat (Real.log 2)
  PhysicsOfConsciousness.Chain.t5_refiningEnergy 3
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.t5_refiningEnergy_tendsto
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
```

A.106 chain-active-hypotheses-jointly-satisfiable

PhysicsOfConsciousness.Chain.chain_active_hypotheses_jointly_satisfiable
[Source, line 1446](#); 5 nested tactic records.

PhysicsOfConsciousness.Chain.UnifiedSelf PhysicsOfConsciousness.Examples.cortexReflexive

Direct project constants in the proof term:

PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.Chain.IsUnifiedBy
PhysicsOfConsciousness.Chain.Unity._proof_1
PhysicsOfConsciousness.Chain.chain_active
PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
PhysicsOfConsciousness.Chain.e67_three_one
PhysicsOfConsciousness.Chain.t5_e12_doubleWell
PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister
PhysicsOfConsciousness.Chain.t5_refiningEnergy
PhysicsOfConsciousness.Chain.thermalAgency_e34Active
PhysicsOfConsciousness.Chain.thermalAgency_e45Active
PhysicsOfConsciousness.DynamicalVacuum
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.cortexCover
PhysicsOfConsciousness.Examples.cortexCover_reachedByRelaxation_three
PhysicsOfConsciousness.Examples.cortexPredict_fixed
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate
PhysicsOfConsciousness.Examples.cortexReflexive
PhysicsOfConsciousness.Examples.cortexState
PhysicsOfConsciousness.Examples.cortexTau
PhysicsOfConsciousness.Examples.cortexTau_pos
PhysicsOfConsciousness.Examples.gsComplete
PhysicsOfConsciousness.Examples.gsMetric
PhysicsOfConsciousness.Examples.instBorelSpaceSite
PhysicsOfConsciousness.Examples.instTriangulatedManifoldCarrierCortex
PhysicsOfConsciousness.Examples.kink
PhysicsOfConsciousness.Examples.wellV
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.resonanceRate

A.107 registerbudget-e34active

PhysicsOfConsciousness.Chain.registerBudget_e34Active
[Source, line 1470](#); 0 nested tactic records.

PhysicsOfConsciousness.Chain.E34Active PhysicsOfConsciousness.Examples.RegisterBudget.cycle.update
(PhysicsOfConsciousness.Examples.RegisterBudget.cycle.step 0)
(PhysicsOfConsciousness.Examples.RegisterBudget.cycle.heat 0)

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.e34Active_of_ledger
PhysicsOfConsciousness.Examples.RegisterBudget.act_others_compressive
PhysicsOfConsciousness.Examples.RegisterBudget.cycle
PhysicsOfConsciousness.Examples.boolThermo

```

A.108 registerbudget-activebound

PhysicsOfConsciousness.Chain.registerBudget_activeBound
[Source, line 1477](#); 0 nested tactic records.

```

PhysicsOfConsciousness.Chain.ActiveBound
  (PhysicsOfConsciousness.Examples.RegisterBudget.cycle.step 0) 1
  (PhysicsOfConsciousness.Examples.RegisterBudget.cycle.heat 0) (Real.log 2)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.activeBound_of_e34Active
PhysicsOfConsciousness.Chain.dissipates_of_not_surjective
PhysicsOfConsciousness.Chain.registerBudget_e34Active
PhysicsOfConsciousness.Chain.witness_not_surjective
PhysicsOfConsciousness.Examples.RegisterBudget.cycle
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.boolThermo
PhysicsOfConsciousness.RegisterLedger.heat
PhysicsOfConsciousness.RegisterLedger.step

```

A.109 chain-active-budget-jointly-satisfiable

PhysicsOfConsciousness.Chain.chain_active_budget_jointly_satisfiable
[Source, line 1489](#); 5 nested tactic records.

```

PhysicsOfConsciousness.Chain.UnifiedSelf PhysicsOfConsciousness.Examples.cortexReflexive

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.ActiveBound
PhysicsOfConsciousness.Chain.CoarseGrains
PhysicsOfConsciousness.Chain.Coherent
PhysicsOfConsciousness.Chain.FieldRealizes
PhysicsOfConsciousness.Chain.IsUnifiedBy
PhysicsOfConsciousness.Chain.Unity._proof_1
PhysicsOfConsciousness.Chain.chain_active
PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
PhysicsOfConsciousness.Chain.e67_three_one
PhysicsOfConsciousness.Chain.registerBudget_e34Active
PhysicsOfConsciousness.Chain.t5_e12_doubleWell
PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister
PhysicsOfConsciousness.Chain.t5_refiningEnergy
PhysicsOfConsciousness.Chain.t5_refiningEnergy_tendsto
PhysicsOfConsciousness.DynamicalVacuum
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.RegisterBudget.cycle
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.boolThermo
PhysicsOfConsciousness.Examples.cortexCover
PhysicsOfConsciousness.Examples.cortexCover_reachedByRelaxation_three
PhysicsOfConsciousness.Examples.cortexPredict_fixed
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate
PhysicsOfConsciousness.Examples.cortexReflexive
PhysicsOfConsciousness.Examples.cortexState
PhysicsOfConsciousness.Examples.cortexTau

```

```

PhysicsOfConsciousness.Examples.cortexTau_pos
PhysicsOfConsciousness.Examples.gsComplete
PhysicsOfConsciousness.Examples.gsMetric
PhysicsOfConsciousness.Examples.instBorelSpaceSite
PhysicsOfConsciousness.Examples.instTriangulatedManifoldCarrierCortex
PhysicsOfConsciousness.Examples.kink
PhysicsOfConsciousness.Examples.wellV
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.RegisterLedger.heat
PhysicsOfConsciousness.RegisterLedger.step
PhysicsOfConsciousness.StatisticalMechanics.toThermodynamics
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.Thermodynamics.temperature
PhysicsOfConsciousness.heat_dissipation
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.resonanceRate

```

A.110 actuated-activebound

PhysicsOfConsciousness.Chain.actuated_activeBound
[Source, line 1508](#); 0 nested tactic records.

```

PhysicsOfConsciousness.Chain.ActiveBound
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.step
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.temperature
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.heat (Real.log 2)

```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.thermalAgency_activeBound
```

A.111 actuated-budget-cap

PhysicsOfConsciousness.Chain.actuated_budget_cap
[Source, line 1515](#); 19 nested tactic records.

```

PhysicsOfConsciousness.Examples.ActuatedField.normalized.continuumLimit ≤
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.gain * Real.log 2 *
  PhysicsOfConsciousness.Examples.ActuatedField.profileMass

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.ActuatedCoupling.base
PhysicsOfConsciousness.ActuatedCoupling.continuumLimit
PhysicsOfConsciousness.ActuatedCoupling.gain
PhysicsOfConsciousness.ActuatedCoupling.region
PhysicsOfConsciousness.ActuatedCoupling.temperature
PhysicsOfConsciousness.ActuatedCoupling.volume
PhysicsOfConsciousness.Chain.actuated_activeBound
PhysicsOfConsciousness.Chain.actuated_limit_le_budget
PhysicsOfConsciousness.Examples.ActuatedField.normalized
PhysicsOfConsciousness.Examples.ActuatedField.profileMass
PhysicsOfConsciousness.Examples.ActuatedField.profileMass_pos

```

A.112 actuated-no-gain-free-budget-cap

PhysicsOfConsciousness.Chain.actuated_no_gain_free_budget_cap
[Source, line 1525](#); 58 nested tactic records.

```
∃ A,  
  PhysicsOfConsciousness.Chain.ActiveBound A.step A.temperature A.heat (Real.log 2) ∧  
    Real.log 2 < A.continuumLimit
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ActuatedCoupling  
PhysicsOfConsciousness.ActuatedCoupling.continuumLimit  
PhysicsOfConsciousness.ActuatedCoupling.heat  
PhysicsOfConsciousness.ActuatedCoupling.step  
PhysicsOfConsciousness.ActuatedCoupling.temperature  
PhysicsOfConsciousness.Chain.ActiveBound  
PhysicsOfConsciousness.Chain.actuated_activeBound  
PhysicsOfConsciousness.Examples.ActuatedField.normalized  
PhysicsOfConsciousness.Examples.ActuatedField.normalized_limit
```

A.113 actuated-wrong-limit-rejected

PhysicsOfConsciousness.Chain.actuated_wrong_limit_rejected
[Source, line 1537](#); 47 nested tactic records.

```
¬PhysicsOfConsciousness.Chain.E45Active  
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.step 1  
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.heat (Real.log 2)  
  PhysicsOfConsciousness.Examples.ActuatedField.normalized.energy 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ActuatedCoupling.actuated_coarseGrains  
PhysicsOfConsciousness.ActuatedCoupling.continuumLimit  
PhysicsOfConsciousness.ActuatedCoupling.energy  
PhysicsOfConsciousness.ActuatedCoupling.heat  
PhysicsOfConsciousness.ActuatedCoupling.step  
PhysicsOfConsciousness.Chain.E45Active  
PhysicsOfConsciousness.Chain.actuated_activeBound  
PhysicsOfConsciousness.Examples.ActuatedField.normalized  
PhysicsOfConsciousness.Examples.ActuatedField.normalized_limit
```

A.114 chain-active-actuated-jointly-satisfiable

PhysicsOfConsciousness.Chain.chain_active_actuated_jointly_satisfiable
[Source, line 1549](#); 28 nested tactic records.

```
PhysicsOfConsciousness.Chain.UnifiedSelf PhysicsOfConsciousness.Examples.cortexReflexive
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.ActuatedCoupling.continuumLimit  
PhysicsOfConsciousness.ActuatedCoupling.energy  
PhysicsOfConsciousness.ActuatedCoupling.heat  
PhysicsOfConsciousness.ActuatedCoupling.step  
PhysicsOfConsciousness.ActuatedCoupling.temperature  
PhysicsOfConsciousness.Chain.CoarseGrains  
PhysicsOfConsciousness.Chain.Coherent  
PhysicsOfConsciousness.Chain.E45Active  
PhysicsOfConsciousness.Chain.FieldRealizes  
PhysicsOfConsciousness.Chain.IsUnifiedBy
```

```

PhysicsOfConsciousness.Chain.Unity._proof_1
PhysicsOfConsciousness.Chain.chain_active
PhysicsOfConsciousness.Chain.cortexMeasureSpace
PhysicsOfConsciousness.Chain.cortexNeuralField
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling
PhysicsOfConsciousness.Chain.e45Active_of_actuatedCoupling
PhysicsOfConsciousness.Chain.e67_three_one
PhysicsOfConsciousness.Chain.t5_e12_doubleWell
PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister
PhysicsOfConsciousness.Chain.thermalAgency_e34Active
PhysicsOfConsciousness.DynamicalVacuum
PhysicsOfConsciousness.Examples.ActuatedField.normalized
PhysicsOfConsciousness.Examples.ActuatedField.normalized_limit
PhysicsOfConsciousness.Examples.Cortex
PhysicsOfConsciousness.Examples.boolStatMech
PhysicsOfConsciousness.Examples.cortexCover
PhysicsOfConsciousness.Examples.cortexCover_reachedByRelaxation_three
PhysicsOfConsciousness.Examples.cortexPredict_fixed
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate
PhysicsOfConsciousness.Examples.cortexReflexive
PhysicsOfConsciousness.Examples.cortexState
PhysicsOfConsciousness.Examples.cortexTau
PhysicsOfConsciousness.Examples.cortexTau_pos
PhysicsOfConsciousness.Examples.gsComplete
PhysicsOfConsciousness.Examples.gsMetric
PhysicsOfConsciousness.Examples.instBorelSpaceSite
PhysicsOfConsciousness.Examples.instTriangulatedManifoldCarrierCortex
PhysicsOfConsciousness.Examples.kink
PhysicsOfConsciousness.Examples.wellV
PhysicsOfConsciousness.GlobalSection
PhysicsOfConsciousness.IsEMFieldCoupling
PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.resonanceRate

```

A.115 microscopic-e45active

PhysicsOfConsciousness.Chain.microscopic_e45Active

[Source, line 1574](#); 6 nested tactic records.

```

PhysicsOfConsciousness.Chain.E45Active PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat (Real.log 2)
    PhysicsOfConsciousness.Examples.MicroscopicCoupling.meshEnergy 3

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Chain.E45Active
PhysicsOfConsciousness.Chain.e45Active_of_localActuator
PhysicsOfConsciousness.Examples.MicroscopicCoupling.installed
PhysicsOfConsciousness.Examples.MicroscopicCoupling.installed_continuumEnergy
PhysicsOfConsciousness.Examples.MicroscopicCoupling.meshEnergy
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
PhysicsOfConsciousness.KernelArrangement.continuumEnergy
PhysicsOfConsciousness.KernelArrangement.energy
PhysicsOfConsciousness.KernelArrangement.law
PhysicsOfConsciousness.ProbDist

```

A.116 microscopic-wrong-limit-rejected

PhysicsOfConsciousness.Chain.microscopic_wrong_limit_rejected

[Source, line 1583](#); 47 nested tactic records.

```
¬PhysicsOfConsciousness.Chain.E45Active PhysicsOfConsciousness.Examples.ThermalAgency.actuation 1
  PhysicsOfConsciousness.Examples.ThermalAgency.heat (Real.log 2)
  PhysicsOfConsciousness.Examples.MicroscopicCoupling.meshEnergy 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E45Active
PhysicsOfConsciousness.Chain.thermalAgency_activeBound
PhysicsOfConsciousness.Examples.MicroscopicCoupling.continuumEnergy
PhysicsOfConsciousness.Examples.MicroscopicCoupling.continuumEnergy_eq
PhysicsOfConsciousness.Examples.MicroscopicCoupling.meshEnergy
PhysicsOfConsciousness.Examples.MicroscopicCoupling.spatial_limit
PhysicsOfConsciousness.Examples.ThermalAgency.actuation
PhysicsOfConsciousness.Examples.ThermalAgency.heat
```

A.117 microscopic-e45

PhysicsOfConsciousness.Chain.microscopic_e45

[Source, line 1595](#); 6 nested tactic records.

```
PhysicsOfConsciousness.Chain.E45 Bool Bool Bool
  PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered.energy 4
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E45
PhysicsOfConsciousness.Chain.e45_of_localActuator
PhysicsOfConsciousness.Examples.MicroscopicCoupling.frozen_law
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered_continuumEnergy
PhysicsOfConsciousness.Examples.frozenSystem
PhysicsOfConsciousness.KernelArrangement.continuumEnergy
PhysicsOfConsciousness.KernelArrangement.energy
```

A.118 microscopic-e45-scrambled

PhysicsOfConsciousness.Chain.microscopic_e45_scrambled

[Source, line 1603](#); 6 nested tactic records.

```
PhysicsOfConsciousness.Chain.E45 Bool Bool Bool
  PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered.energy 4
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E45
PhysicsOfConsciousness.Chain.e45_of_localActuator
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered_continuumEnergy
PhysicsOfConsciousness.Examples.MicroscopicCoupling.scrambled_law
PhysicsOfConsciousness.Examples.scrambledSystem
PhysicsOfConsciousness.KernelArrangement.continuumEnergy
PhysicsOfConsciousness.KernelArrangement.energy
```

A.119 microscopic-passive-wrong-limit-rejected

PhysicsOfConsciousness.Chain.microscopic_passive_wrong_limit_rejected
[Source, line 1611](#); 19 nested tactic records.

```
¬PhysicsOfConsciousness.Chain.E45 Bool Bool Bool  
  PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered.energy 0
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.E45  
PhysicsOfConsciousness.Chain.predictiveBound_of_nonempty  
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered  
PhysicsOfConsciousness.Examples.MicroscopicCoupling.remembered_limit  
PhysicsOfConsciousness.Examples.frozenSystem  
PhysicsOfConsciousness.KernelArrangement.energy  
PhysicsOfConsciousness.PredictiveDissipation
```

A.120 chain-active-microscopic-jointly-satisfiable

PhysicsOfConsciousness.Chain.chain_active_microscopic_jointly_satisfiable
[Source, line 1628](#); 5 nested tactic records.

```
PhysicsOfConsciousness.Chain.UnifiedSelf PhysicsOfConsciousness.Examples.cortexReflexive
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Chain.CoarseGrains  
PhysicsOfConsciousness.Chain.Coherent  
PhysicsOfConsciousness.Chain.FieldRealizes  
PhysicsOfConsciousness.Chain.IsUnifiedBy  
PhysicsOfConsciousness.Chain.Unity._proof_1  
PhysicsOfConsciousness.Chain.chain_active  
PhysicsOfConsciousness.Chain.cortexMeasureSpace  
PhysicsOfConsciousness.Chain.cortexNeuralField  
PhysicsOfConsciousness.Chain.cortexNeuralField_isEMFieldCoupling  
PhysicsOfConsciousness.Chain.e67_three_one  
PhysicsOfConsciousness.Chain.microscopic_e45Active  
PhysicsOfConsciousness.Chain.t5_e12_doubleWell  
PhysicsOfConsciousness.Chain.t5_e23_absorbingRegister  
PhysicsOfConsciousness.Chain.thermalAgency_e34Active  
PhysicsOfConsciousness.DynamicalVacuum  
PhysicsOfConsciousness.Examples.Cortex  
PhysicsOfConsciousness.Examples.MicroscopicCoupling.meshEnergy  
PhysicsOfConsciousness.Examples.ThermalAgency.actuation  
PhysicsOfConsciousness.Examples.ThermalAgency.heat  
PhysicsOfConsciousness.Examples.boolStatMech  
PhysicsOfConsciousness.Examples.cortexCover  
PhysicsOfConsciousness.Examples.cortexCover_reachedByRelaxation_three  
PhysicsOfConsciousness.Examples.cortexPredict_fixed  
PhysicsOfConsciousness.Examples.cortexPredict_lipschitz_rate  
PhysicsOfConsciousness.Examples.cortexReflexive  
PhysicsOfConsciousness.Examples.cortexState  
PhysicsOfConsciousness.Examples.cortexTau  
PhysicsOfConsciousness.Examples.cortexTau_pos  
PhysicsOfConsciousness.Examples.gsComplete  
PhysicsOfConsciousness.Examples.gsMetric  
PhysicsOfConsciousness.Examples.instBorelSpaceSite  
PhysicsOfConsciousness.Examples.instTriangulatedManifoldCarrierCortex  
PhysicsOfConsciousness.Examples.kink  
PhysicsOfConsciousness.Examples.wellV  
PhysicsOfConsciousness.GlobalSection  
PhysicsOfConsciousness.IsEMFieldCoupling
```

```

PhysicsOfConsciousness.LocalSectionSynchronization.I
PhysicsOfConsciousness.LocalSectionSynchronization.cover
PhysicsOfConsciousness.ReflexiveBoundary.predict
PhysicsOfConsciousness.ThermodynamicCover.IsReachedByRelaxation
PhysicsOfConsciousness.ThermodynamicCover.toLocalSectionSynchronization
PhysicsOfConsciousness.probabilityPresheaf
PhysicsOfConsciousness.resonanceRate

```

A.121 nonconstant-witness

PhysicsOfConsciousness.Examples.richDensity_not_const
[Source, line 100](#); 19 nested tactic records.

```

PhysicsOfConsciousness.Examples.richDensity 0 ≠ PhysicsOfConsciousness.Examples.richDensity Real.pi

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Examples.Site
PhysicsOfConsciousness.Examples.Site.left
PhysicsOfConsciousness.Examples.richDensity

```

A.122 finite-measure-gluing

PhysicsOfConsciousness.SpatialMeasure.finite_glue_unique
[Source, line 19](#); 259 nested tactic records.

```

∀ {X : Type u_1} {I : Type u_2} [inst : MeasurableSpace X] [Fintype I] (U : I → Set X),
  (∀ (i : I), MeasurableSet (U i)) →
    ⋃ i, U i = Set.univ →
      ∀ (μ : I → MeasureTheory.FiniteMeasure X),
        (∀ (i j : I), (↑(μ i)).restrict (U i ∩ U j) = (↑(μ j)).restrict (U i ∩ U j)) →
          ∃! ν, ∀ (i : I), (↑ν).restrict (U i) = (↑(μ i)).restrict (U i)

```

A.123 infinite-measure-obstruction

PhysicsOfConsciousness.SpatialMeasure.no_finite_measure_of_unit_atoms
[Source, line 55](#); 67 nested tactic records.

```

¬∃ μ, ∀ (n : ℕ), ↑μ {n} = 1

```

A.124 measure-witness-extension

PhysicsOfConsciousness.Examples.MeasureGluing.extension_eq
[Source, line 53](#); 0 nested tactic records.

```

∀ (ν : MeasureTheory.FiniteMeasure ℝ),
  (∀ (i : Bool),
    (↑ν).restrict (PhysicsOfConsciousness.Examples.MeasureGluing.patch i) =
      (↑(PhysicsOfConsciousness.Examples.MeasureGluing.localMeasure i)).restrict
        (PhysicsOfConsciousness.Examples.MeasureGluing.patch i)) →
    ν = PhysicsOfConsciousness.Examples.MeasureGluing.glued

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.Examples.MeasureGluing.glued
PhysicsOfConsciousness.Examples.MeasureGluing.glued_restrict
PhysicsOfConsciousness.Examples.MeasureGluing.glued_unique
PhysicsOfConsciousness.Examples.MeasureGluing.localMeasure
PhysicsOfConsciousness.Examples.MeasureGluing.patch

```

A.125 measure-witness-mass

PhysicsOfConsciousness.Examples.MeasureGluIng.glued_mass
[Source, line 58](#); 5 nested tactic records.

```
⌈PhysicsOfConsciousness.Examples.MeasureGluIng.glued Set.univ = 3
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Examples.MeasureGluIng.glued  
PhysicsOfConsciousness.Examples.MeasureGluIng.localMeasure._proof_1
```

A.126 measure-witness-distinct

PhysicsOfConsciousness.Examples.MeasureGluIng.local_measures_differ
[Source, line 61](#); 19 nested tactic records.

```
PhysicsOfConsciousness.Examples.MeasureGluIng.localMeasure false ≠  
PhysicsOfConsciousness.Examples.MeasureGluIng.localMeasure true
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.Examples.MeasureGluIng.localMeasure  
PhysicsOfConsciousness.Examples.MeasureGluIng.localMeasure._proof_1
```

A.127 cell-kernel-integral

PhysicsOfConsciousness.KernelMesh.energy_eq_integral
[Source, line 48](#); 93 nested tactic records.

```
∀ {M : Type u_1} [inst : MeasurableSpace M] (G : PhysicsOfConsciousness.KernelMesh M)  
  (μ : MeasureTheory.Measure M) [MeasureTheory.IsFiniteMeasure μ] (K : M → M → ℝ),  
  G.energy μ K = ∫ (z : M × M), G.kernel K z.1 z.2 ∂μ.prod μ
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.KernelMesh  
PhysicsOfConsciousness.KernelMesh.cell  
PhysicsOfConsciousness.KernelMesh.energy  
PhysicsOfConsciousness.KernelMesh.energy_eq_integral._simp_2  
PhysicsOfConsciousness.KernelMesh.kernel  
PhysicsOfConsciousness.KernelMesh.locate  
PhysicsOfConsciousness.KernelMesh.locate_measurable  
PhysicsOfConsciousness.KernelMesh.sample  
PhysicsOfConsciousness.KernelMesh.size
```

A.128 cell-kernel-limit

PhysicsOfConsciousness.KernelMesh.kernel_tendsto
[Source, line 76](#); 0 nested tactic records.

```
∀ {M : Type u_1} [inst : MeasurableSpace M] [inst_1 : TopologicalSpace M]  
  (G : ℕ → PhysicsOfConsciousness.KernelMesh M) (K : M → M → ℝ),  
  Continuous (Function.uncurry K) →  
  (∀ (x : M), Filter.Tendsto (fun n => (G n).snap x) Filter.atTop (ℕ x)) →  
  ∀ (x y : M), Filter.Tendsto (fun n => (G n).kernel K x y) Filter.atTop (ℕ (K x y))
```

Direct project constants in the proof term:

```
PhysicsOfConsciousness.KernelMesh  
PhysicsOfConsciousness.KernelMesh.snap
```

A.129 cell-energy-limit

PhysicsOfConsciousness.KernelMesh.energy_tendsto

Source, line 86; 46 nested tactic records.

```

V {M : Type u_1} [inst : MeasurableSpace M] [inst_1 : TopologicalSpace M] [CompactSpace M]
  [SecondCountableTopology M] [BorelSpace M] (G : ℕ → PhysicsOfConsciousness.KernelMesh M)
  (μ : MeasureTheory.Measure M) [MeasureTheory.IsFiniteMeasure μ] (K : M → M → ℝ),
Continuous (Function.uncurry K) →
  (∀ (x : M), Filter.Tendsto (fun n => (G n).snap x) Filter.atTop (λ x) x) →
  Filter.Tendsto (fun n => (G n).energy μ K) Filter.atTop
    (λ (f : M → M), K f.1 f.2 ∂μ.prod μ)

```

Direct project constants in the proof term:

```

PhysicsOfConsciousness.KernelMesh
PhysicsOfConsciousness.KernelMesh.energy
PhysicsOfConsciousness.KernelMesh.energy_eq_integral
PhysicsOfConsciousness.KernelMesh.kernel
PhysicsOfConsciousness.KernelMesh.kernel_tendsto
PhysicsOfConsciousness.KernelMesh.snap
PhysicsOfConsciousness.KernelMesh.snap_measurable

```

B Two extracted physical obligations

These projection types are extracted from the structures used by the gluing results. They show the equilibrium and overlap-agreement obligations described in Section 2. The snapshots contain the other encountered structure fields as well. The exporter follows project statement types and parent fields; it does not claim a complete semantic expansion through every definition body or every Mathlib structure.

```

PhysicsOfConsciousness.ThermodynamicCover.thermodynamic_equilibrium
V {X : TopCat} {inst : MeasurableSpace X} {inst_1 : BorelSpace X}
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold X}
  [self : PhysicsOfConsciousness.ThermodynamicCover X]
  (theta : PhysicsOfConsciousness.LocalSectionSynchronization.I X → ℝ),
PhysicsOfConsciousness.kuramoto_potential_dynamic
  { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm := ... }
PhysicsOfConsciousness.LocalSectionSynchronization.phase ≤
PhysicsOfConsciousness.kuramoto_potential_dynamic
  { omega := fun x => 0, A := PhysicsOfConsciousness.ThermodynamicCover.A, symm := ... } theta

PhysicsOfConsciousness.LocalSectionSynchronization.section_agrees_of_phase_eq
V {X : TopCat} {inst : MeasurableSpace X} {inst_1 : BorelSpace X}
  {inst_2 : PhysicsOfConsciousness.TriangulatedManifold X}
  [self : PhysicsOfConsciousness.LocalSectionSynchronization X]
  (i j : PhysicsOfConsciousness.LocalSectionSynchronization.I X),
Real.cos
  (PhysicsOfConsciousness.LocalSectionSynchronization.phase i -
   PhysicsOfConsciousness.LocalSectionSynchronization.phase j) =
  1 →
(CategoryTheory.ConcreteCategory.hom
  ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ...).op))
  (PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section i) =
(CategoryTheory.ConcreteCategory.hom
  ((PhysicsOfConsciousness.probabilityPresheaf X).map (CategoryTheory.homOfLE ...).op))
  (PhysicsOfConsciousness.LocalSectionSynchronization.sync_to_section j)

```